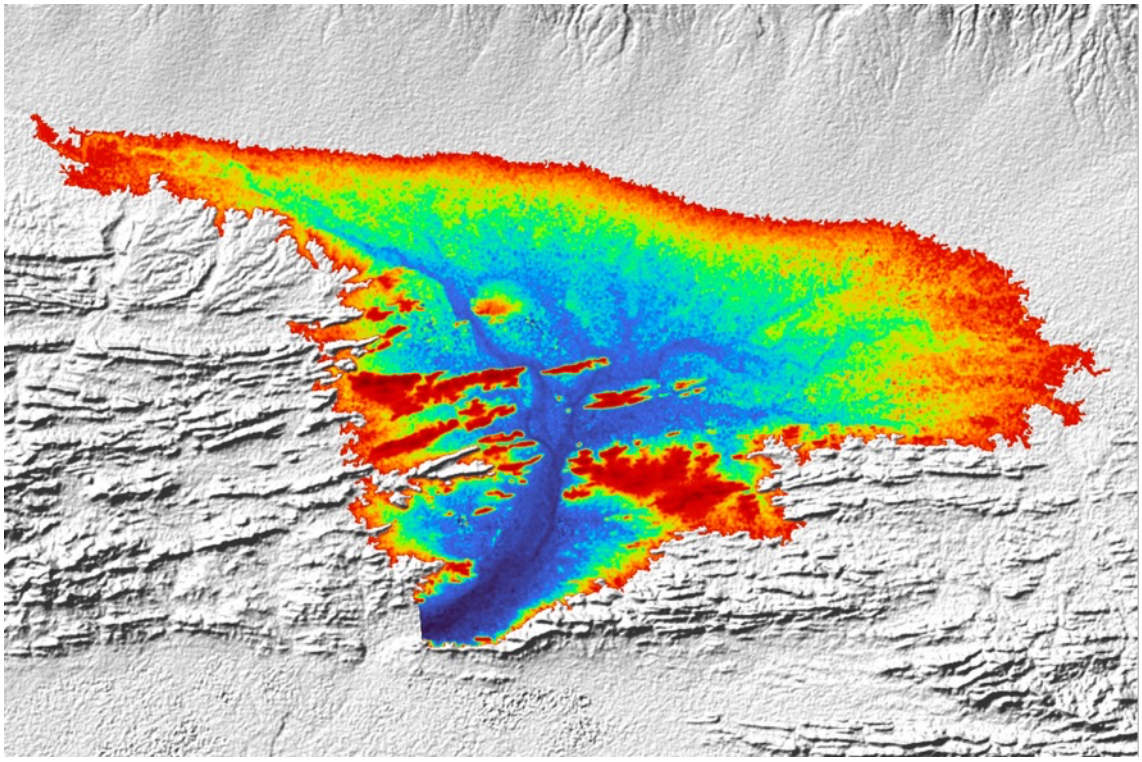
QGIS for Hydrology



Reservoir Elevation-Area-Capacity Curves & Reservoir Monitoring V1

by Andreas de Jong
July 2025

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1 Introduction

QGIS has many powerful functions which we can use for hydrological studies of river basins. There are many free online datasets available for precipitation, evapotranspiration, land use etc., but you will need to first create a "cookie cutter" to extract the relevant data of e.g., a catchment area. This is best done using QGIS.

Reservoir design and monitoring requires Elevation-Area-Capacity curves. These can be calculated in QGIS, and when combined with water area and/or reservoir radar altimetry data we can monitor the volume of water in a reservoir remotely.

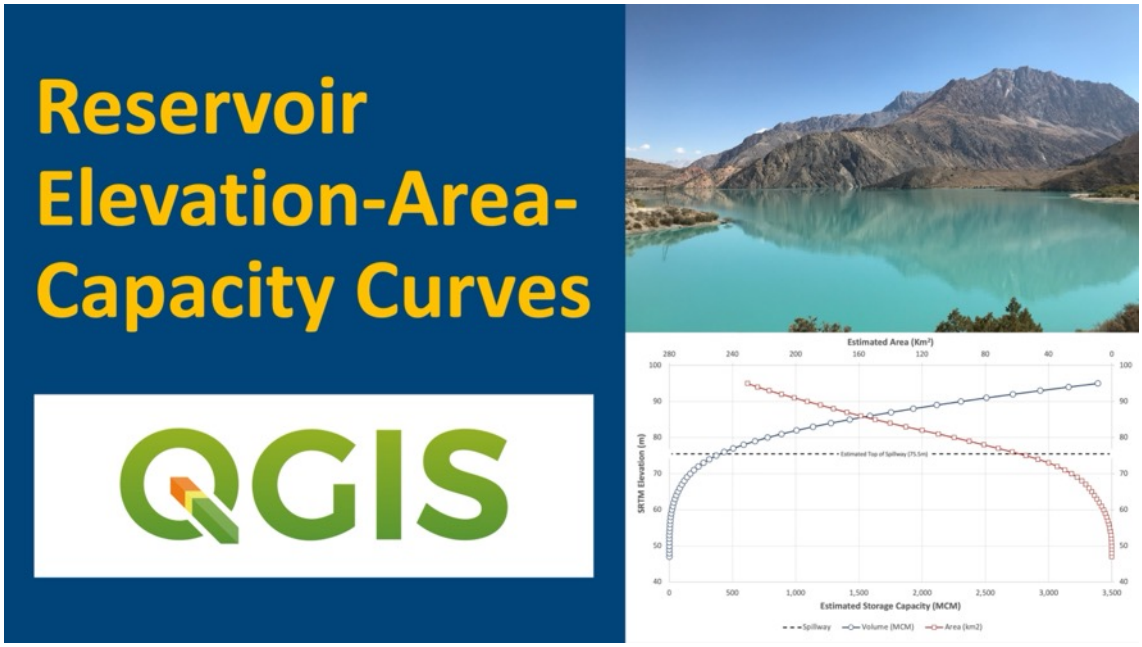
It is assumed that you have successfully completed the Introduction to Q-GIS course, and QGIS in Hydrology – Delineation of Catchments.

1.1 Purpose of this manual

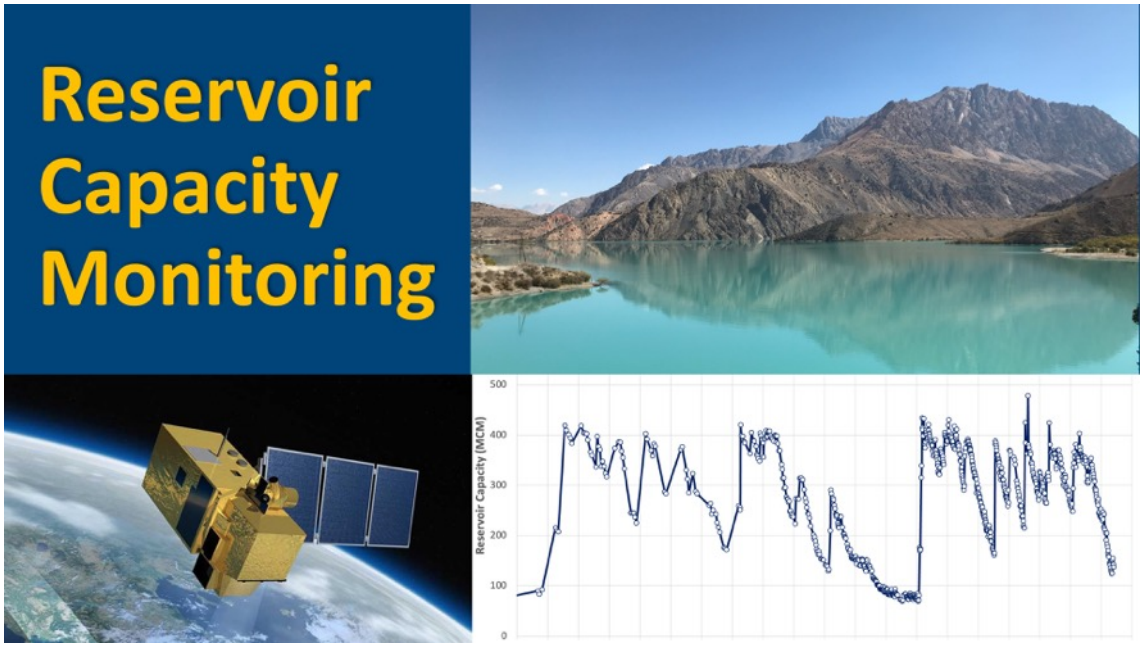
The purpose of this manual is to show you how to derive the Elevation-Area-Capacity curves of reservoirs & use them for monitoring reservoir capacity.

1.2 Video

This manual accompanies two YouTube videos. The first is called **QGIS for Hydrology - Reservoir Elevation-Area-Capacity Curves:** https://youtu.be/InesQRzeqcU which describes the methodology for calculating reservoir curves



The second video is **Reservoir Capacity Monitoring using the Global Water Monitor dataset**: https://youtu.be/-0BGnorZgMg which describes how to download free reservoir monitoring data and display it in the reservoir template.



1.3 Disclaimer

The information and views set out in this manual are those of the author and do not necessarily reflect the official opinion of any institution or government. Neither the author nor any institution or government or any person acting on their behalf may be held responsible for the use which may be made of the information contained therein.

Further comments and suggestions are welcome, and can be submitted to:
Andreas de Jong at info@geosearch.co.uk

1.4 QGIS

QGIS (formally Quantum GIS) is a free, open-source software. You can use it in any way you want, and the maps are yours to keep or distribute. The only thing you need to be careful of is to respect the terms of use of the data providers, if the data is not your own. This usually means acknowledging the source of the data, even if you downloaded it from the internet. QGIS is available for Windows, Mac OS or Linux & you can access it at: https://qgis.org

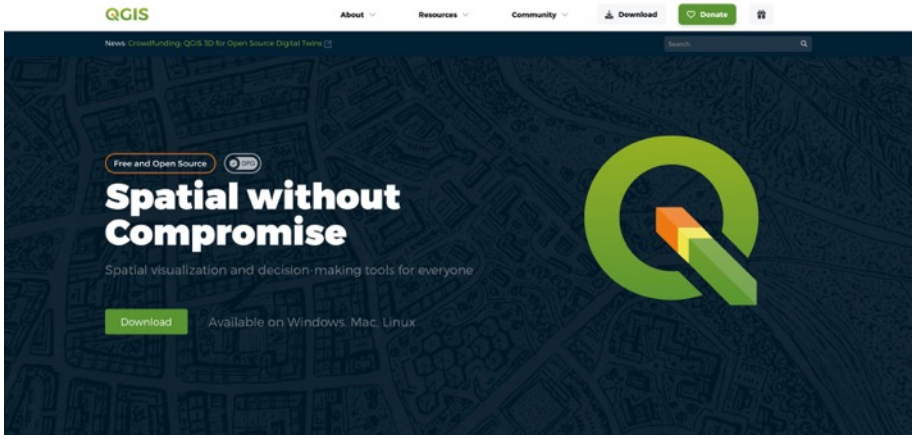


Figure 1: The QGIS Website

1.5 Online QGIS Training Resources

As QGIS is open source rather than commercial it is often difficult to find training material. Here are some useful YouTube Channels:



http://www.youtube.com/@KlasKarlsson



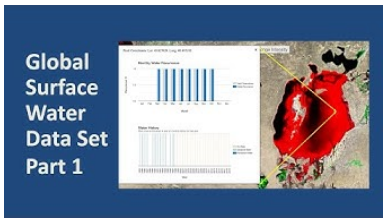
www.youtube.com/@HansvanderKwast



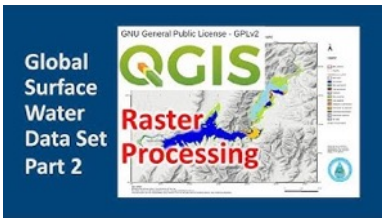
www.youtube.com/@GeosearchInternational

1.6 Library of QGIS Videos

The following videos are currently available on the QGIS 101 playlist of the **GeosearchInternational** YouTube Channel:



https://youtu.be/pOLOdbShNVA



https://youtu.be/ybo_w3SDIbI



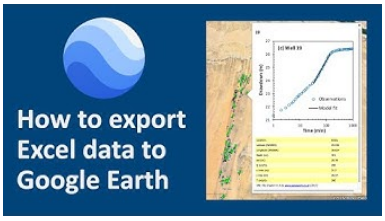
https://youtu.be/TqymUhRWz00



https://youtu.be/uW3XpIzUGcw



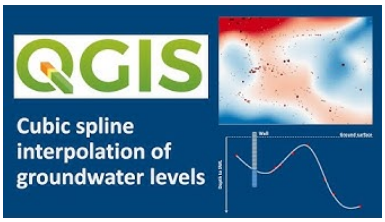
https://youtu.be/OE4IcagdIos



https://youtu.be/4SKMB3v90tE



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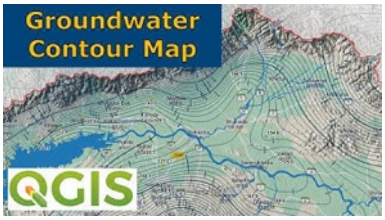
https://youtu.be/_Oe7Uyogdfg



https://youtu.be/rPrC3qOtRAk



https://youtu.be/SU3csjbq5lA



https://youtu.be/Ib8MQ68xemY



https://youtu.be/ceBOzPtQvms



https://youtu.be/9ETEGlIYgis



https://youtu.be/nOAFkmFCO5s

2 The Shuttle Radar Topography (SRTM) Mission & What you need to know about Elevation

Reservoir Elevation-Area-Capacity Curves show the relationship between the water level, the reservoir area and the reservoir capacity. We can calculate these curves in QGIS using the SRTM30 DEM. As this was flown in February 2000, it can be used for any reservoir built after February 2000. For older reservoirs, you can only get the curves above the water level of the reservoir when the SRTM was acquired.

2.1 Key facts about the SRTM Mission

2.1.1 SRTM Acquisition date

The SRTM mission was flown from **February 11, 2000 – February 22, 2000.** Check when your dam was built.

- **Dam built after February 2000:** you can follow the procedures outlined in this manual.
- **Dam built before February 2000:** you cannot get any elevation data for the areas covered by water during February 2000. However, it may still give you useful information, especially if the datum used during dam construction is not a standard one. Please see the procedure on merging reservoir curves from SRTM and dam construction documents explained in my video on the Kakai Dam, Afghanistan.



https://youtu.be/SU3csjbq5lA

2.1.2 SRTM Resolution

The raster resolution is **one arcsecond** which is about **30 m** along the equator. Note that this distance will decrease as we get further away from the equator. It is therefore essential to use only projected SRTM30m elevation grids, if we want to carry out any calculations such as the ones needed for dam curves.

2.1.3 SRTM Elevation

The original SRTM elevations were calculated relative to the **WGS84 ellipsoid** and then the **EGM96 geoid** separation values were added to convert to heights relative to the geoid for all the released products.

Note that the **current Earth Gravitational Model is EGM2008**, so be careful when comparing SRTM with satellite altimetry data.

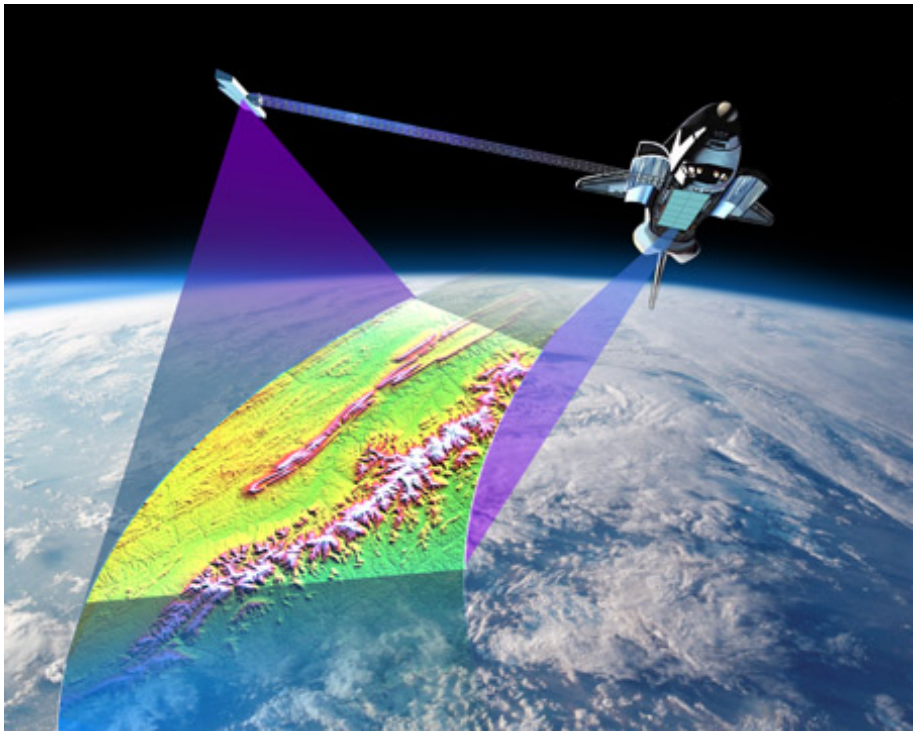


Figure 2: The SRTM mission

Image credit: www.arcgis.com

2.2 Summary of the Geoid, Ellipsoid & Elevation

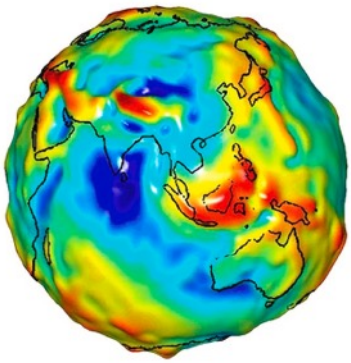
2.2.1 Geoid

The geoid is a real physical surface based on Earth's gravity. It's often used as the "zero" reference for measuring elevations, like "sea level" but extending even under landmasses.

Imagine the Earth covered entirely by water, and this water is perfectly still, without any winds or currents. This imaginary, lumpy, wavy surface is called the geoid.

Why is it lumpy? Because the Earth's gravity isn't perfectly uniform. Denser areas (like mountains beneath the surface) have stronger gravity, pulling the water towards them and creating "bumps" on this imaginary ocean. Less dense areas (like deep ocean trenches) have weaker gravity, causing "dips."

The Indian Ocean Geoid Low (IOGL) is the largest and most pronounced "dip" in the global geoid, located south of the Indian subcontinent, near Sri Lanka. The IOGL represents the lowest geoid anomaly on Earth, with the geoid surface being more than 100 meters lower than the global average (or the reference ellipsoid) in its deepest part. This means that if the ocean were perfectly calm and free of tides and currents, the sea level in that region would literally be over 100 meters lower than in other parts of the world.



2.2.2 Ellipsoid

The ellipsoid is a simplified mathematical model of the Earth's shape. It is useful for calculating positions (latitude and longitude) because it's smooth and predictable.

Imagine a perfectly smooth, mathematical, slightly flattened sphere. This is an ellipsoid. Think of it like a perfectly squashed ball, flatter at the poles and bulging at the equator.

Why is it used? Because the Earth, due to its rotation, isn't a perfect sphere; it's an oblate spheroid (a type of ellipsoid). It's a much simpler shape to describe with mathematical equations than the lumpy geoid.

2.2.3 Elevation

Elevation is simply the height of a point on the Earth's surface above a specific reference level. It is the "real-world" height we often talk about when referring to mountains, cities, or anything on land. Think of it like this: if you are standing on a mountain, your elevation is how high that point is above a standard "zero" level, typically mean sea level.

2.2.4 Ellipsoid Height (also called Geodetic Height)

This is the height of a point on the Earth's surface measured straight up or down from the mathematical ellipsoid. It does not directly account for the local variations in Earth's gravity. GPS devices typically give you an ellipsoid height. They calculate your position relative to this smooth, mathematical model of the Earth.

2.2.5 Geoidal Height (also called Geoid Undulation or Geoid Separation)

This is the distance between the ellipsoid and the geoid at a particular location.
Why is it important? Because the ellipsoid is a smooth model and the geoid is a lumpy physical surface, they don't always perfectly align. The geoidal height tells you how much the geoid "undulates" (goes up or down) relative to the ellipsoid at any given spot.

Relationship to elevation: The geoidal height is crucial for converting between ellipsoid heights (from GPS) and the more familiar "elevation" above sea level (which is relative to the geoid).

2.2.6 Summary

Elevation (Orthometric Height) = Ellipsoid Height - Geoidal Height

So, your GPS might tell you an ellipsoid height, but to get your "true" elevation above sea level, you need to subtract (or add, depending on whether the geoid is above or below the ellipsoid) the geoidal height for your location.

This correction is often done automatically by mapping software and GPS receivers to give you the familiar "elevation."

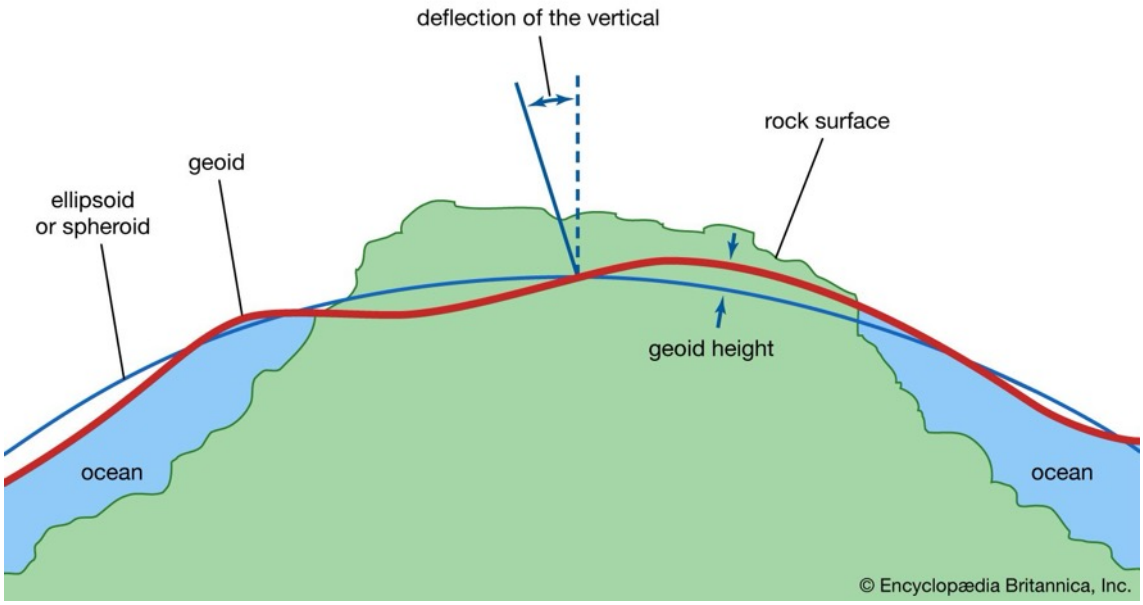


Figure 3: Geoid, ellipsoid, land and ocean surfaces

Image credit: https://cdn.britannica.com/67/467-050-A9D84A6E/Deflection-vertical-geoid-spheroid.jpg

2.3 Important things to remember

1. The SRTM mission was flown in **February 2000**, so it describes the earth's surface at that time. Areas covered by water reflect radar, so SRTM cannot look beneath the water surface.
2. If your reservoir was filled before February 2000 the methodology described in this manual may still be useful, although you will need to find other data sources for the topography of the land surface below the reservoir level at that time.
3. The SRTM grid is in geographic WGS84 coordinates and needs to be converted to a **projected coordinate system**, with a grid size of 30m before any calculations can be carried out.
4. The original SRTM elevations are relative to the **EGM96 geoid**. You will need to correct for this when comparing the data to other coordinate systems.
5. GPS coordinates give elevations relative to the WGS84 ellipsoid, so they cannot be compared with SRTM elevations without correcting for the EGM96 geoidal height at that location.
6. Standard **GPS elevations are not very accurate**. The z-error = 3 x the x-y-error. This means that if a standard GPS position has an accuracy of 4 metres, expect 4 x 3 = 12 metres error in the GPS elevation.

3 Calculate Reservoir Elevation-Area-Capacity Curves in QGIS

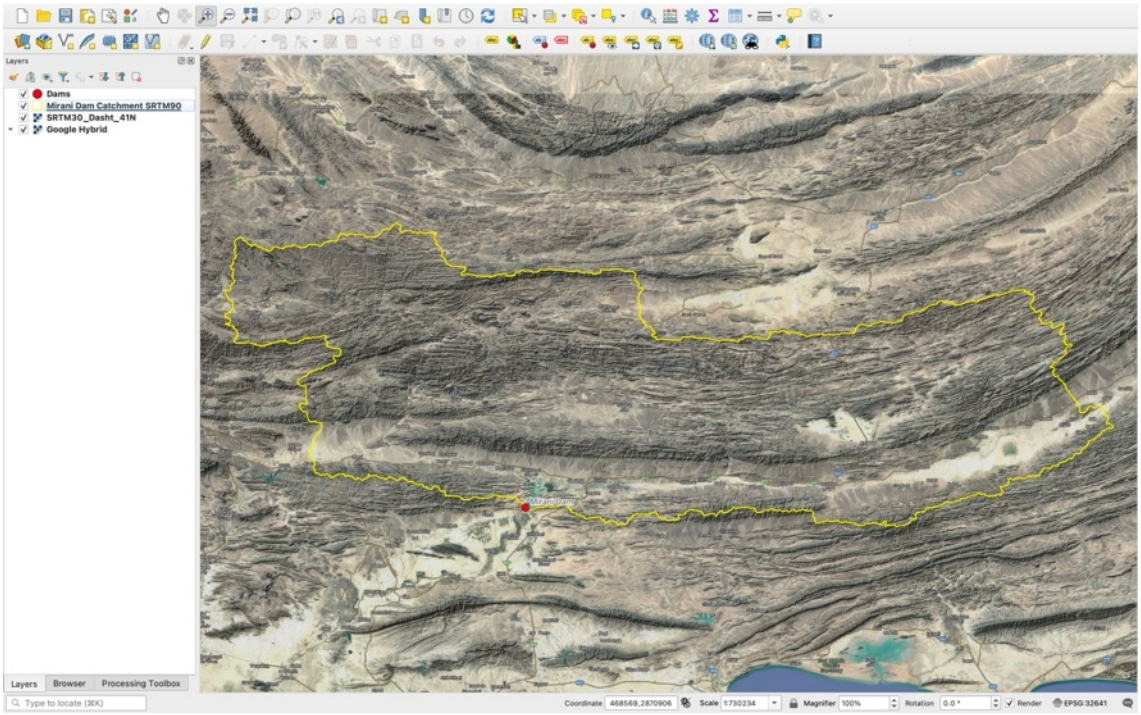
Reservoir Elevation-Area-Capacity Curves show the relationship between the water level, the reservoir area and the reservoir capacity. We can calculate these curves in QGIS using the SRTM30 DEM. As this was flown in February 2000, it can be used for any reservoir built after February 2000. For older reservoirs, you can only get the curves above the water level of the reservoir when the SRTM was acquired.

***It is important that you use a DEM in a projected coordinate system such as UTM**.*

3.1 Set up the GIS

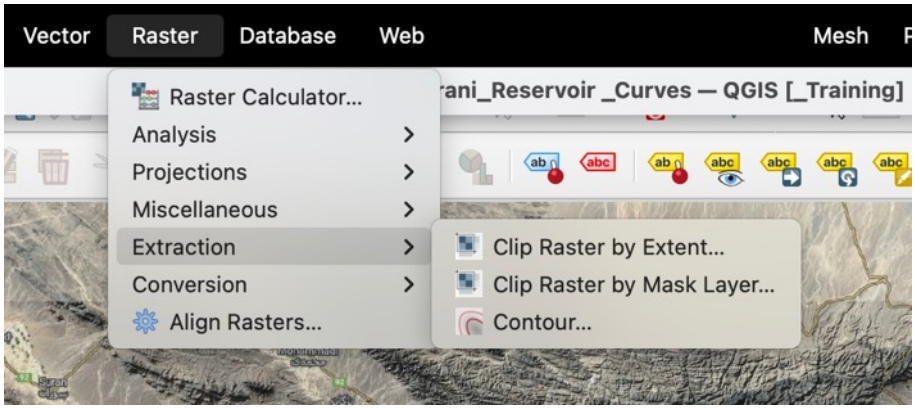
Open QGIS and load the following layers:

- Google Hybrid as a background image
- SRTM30 digital elevation model in projected coordinates
- Location of dam site
- Catchment polygon of dam site

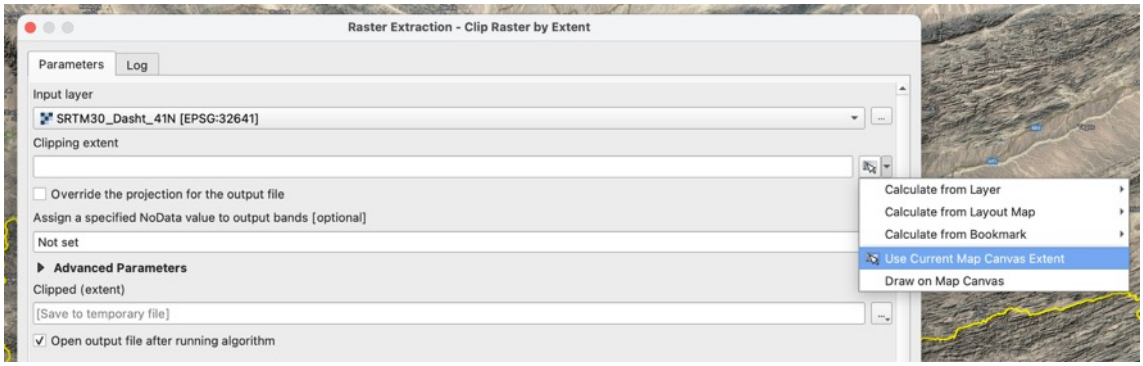


3.2 Extract SRTM30 over the reservoir area

- Zoom to the projected SRTM30 DEM layer which covers the area around the reservoir. It is better to include a much larger area than is absolutely needed as it is difficult to know what the maximum possible flooded area of the reservoir is.
- **Raster | Extraction | Clip Raster by Extent…** Make sure that the area covers all potentially flooded areas of the reservoir.

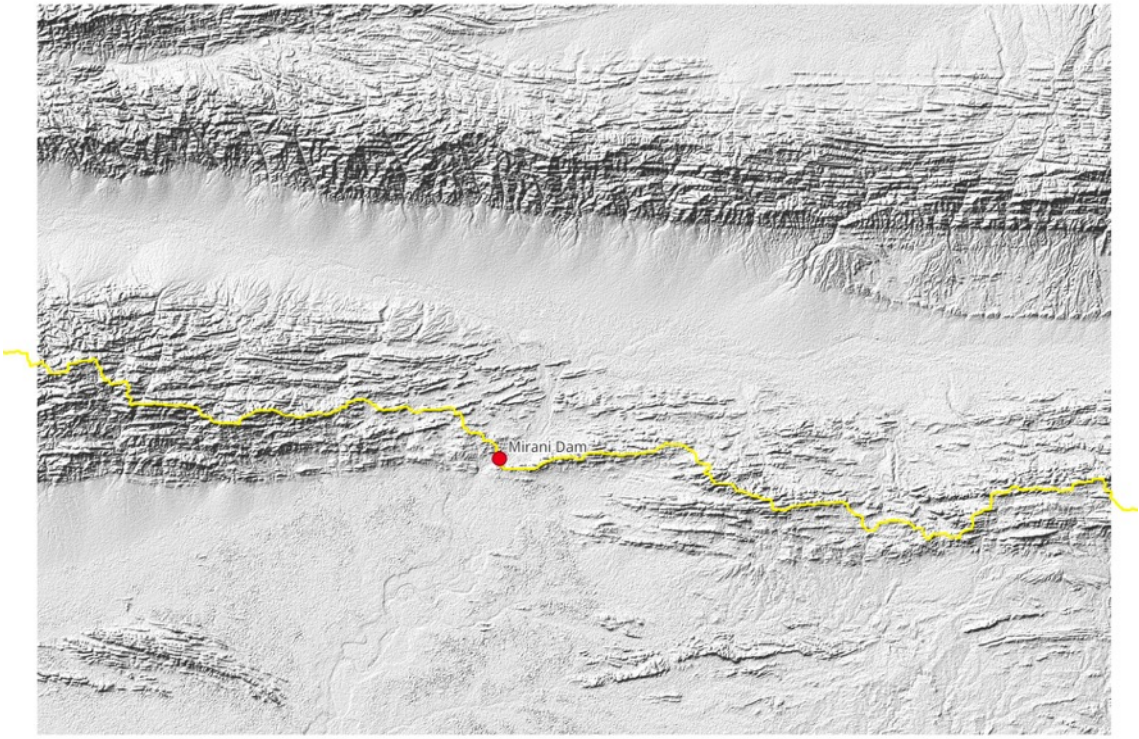


- Clipping Extent: Select **Current Map Canvas Extent**
- Press **Run**



A temporary layer called *Clipped (extent)* will be produced.

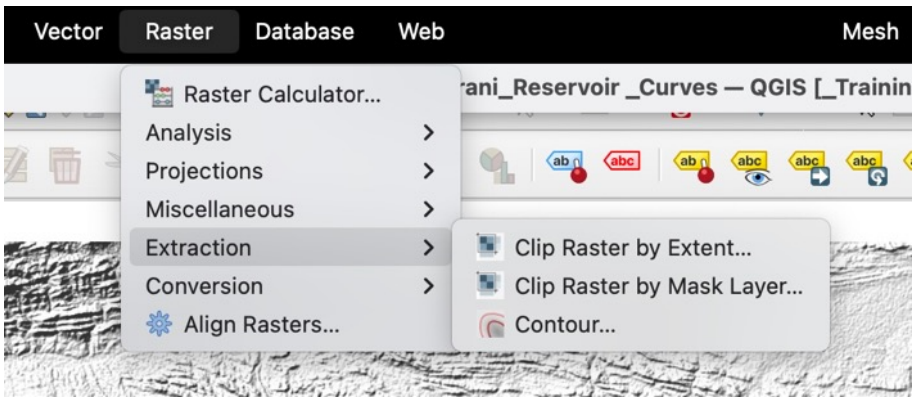
- Copy the style across from the SRTM30 layer**: Right-click | Styles | Copy Styles | All Style Categories**
- Select Clipped (extent): **Right-click | Styles | Paste Styles | All Style Categories**



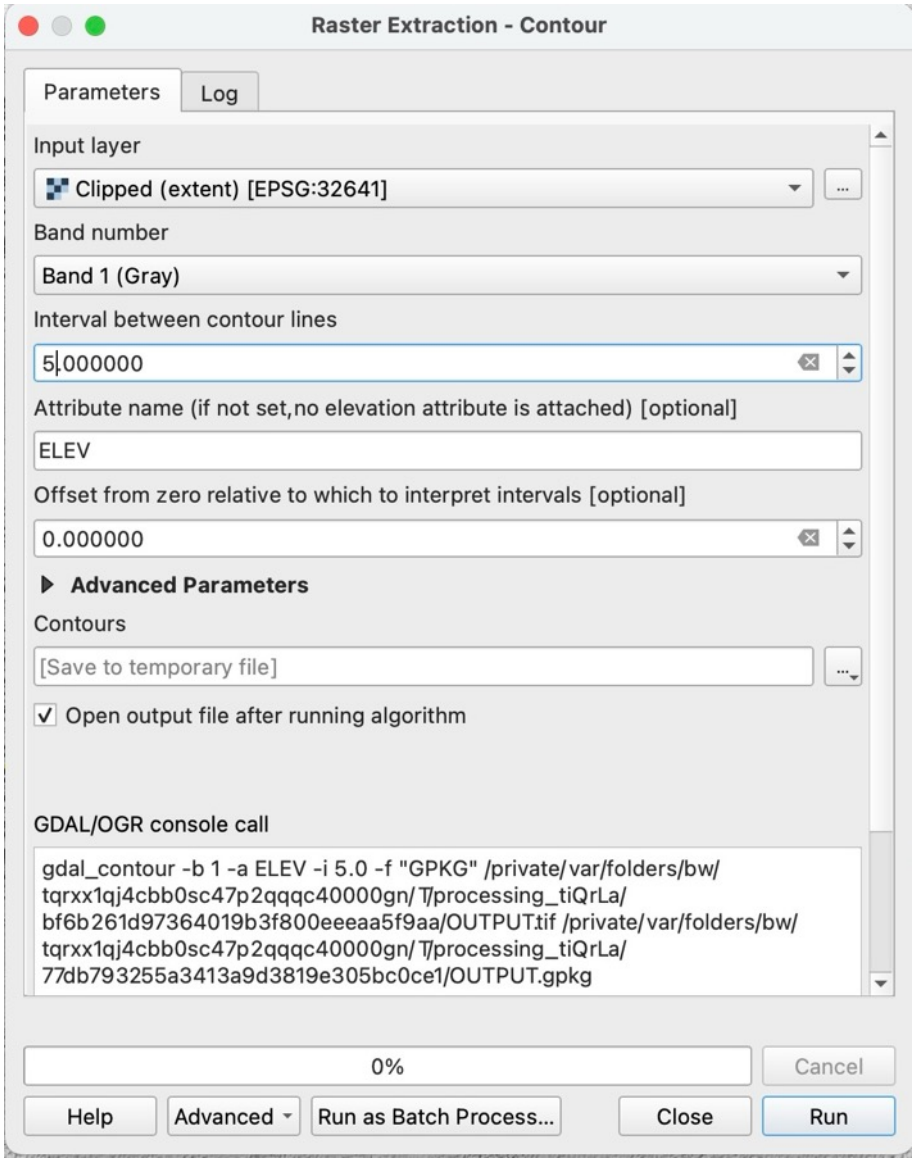
3.3 Create Contours from SRTM30

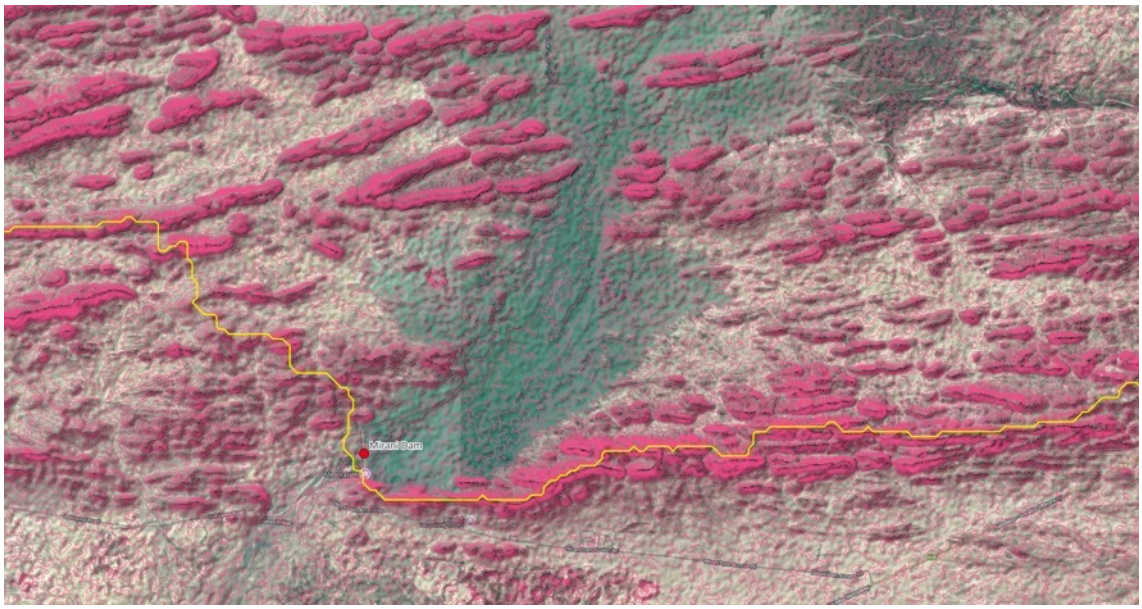
We will use a contour to delineate the maximum flooded area. This should be as a minimum the height of the dam, plus the maximum height of water over the spillway during the maximum design flood. However, it could be even higher, if in future it is decided to raise the height of the dam.

- Select **Raster | Extraction | Contour…**



- **Input layer:** SRTM30 UTM41N digital elevation model
- **Interval between contour lines:** 5m
- **Contours:** [Save to temporary file] to see what it looks like or save directly to a GeoPackage.
- Press **Run**

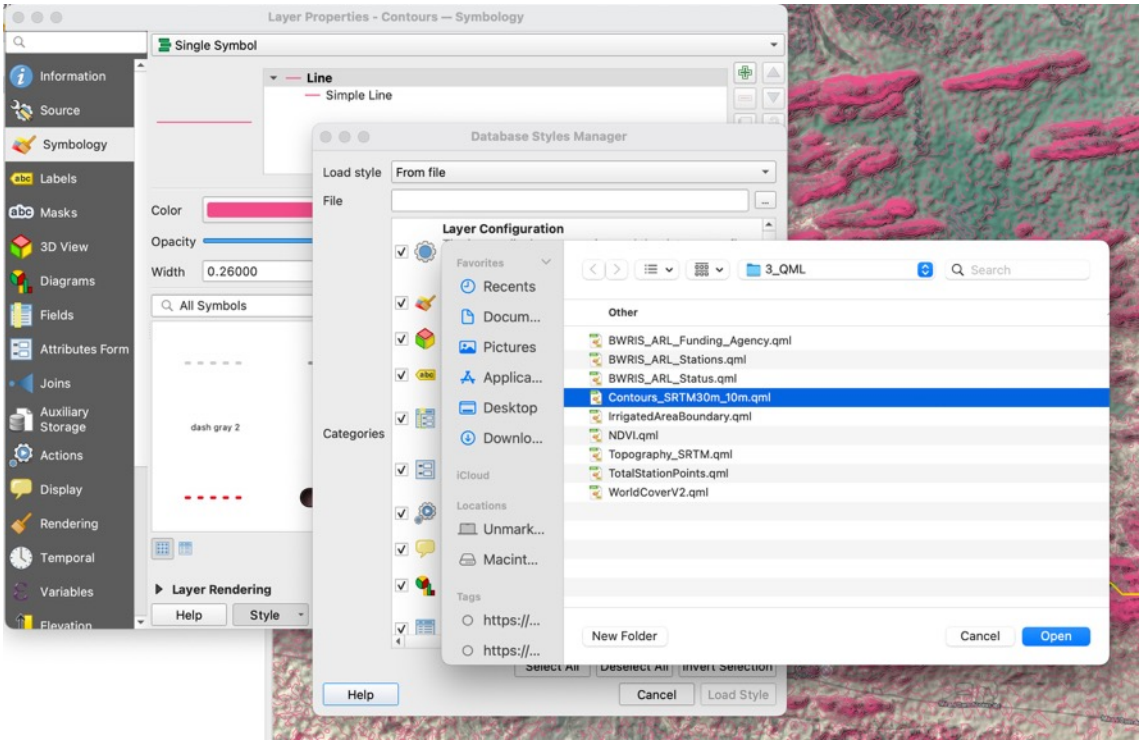

Raster Extraction - Contour
Parameters
Log
Input layer
Clipped (extent) [EPSG:32641]
Band number
Band 1 (Gray)
Interval between contour lines
5.000000
Attribute name (if not set,no elevation attribute is attached) [optional]
ELEV
Offset from zero relative to which to interpret intervals [optional]
0.000000
Advanced Parameters
Contours
[Save to temporary file]
Open output file after running algorithm
GDAL/OGR console call
gdal_contour -b 1 -a ELEV -i 5.0 -f "GPKG" /private/var/folders/bw/tqrxx1qj4cbb0sc47p2qqqc40000gn/T/processing_tiQrLa/bf6b261d97364019b3f800eeeaa5f9aa/OUTPUT.tif /private/var/folders/bw/tqrxx1qj4cbb0sc47p2qqqc40000gn/T/processing_tiQrLa/77db793255a3413a9d3819e305bc0ce1/OUTPUT.gpkg
0%
Cancel
Help
Advanced
Run as Batch Process...
Close
Run



3.4 Delineate the maximum flooded area from a SRTM30 contour

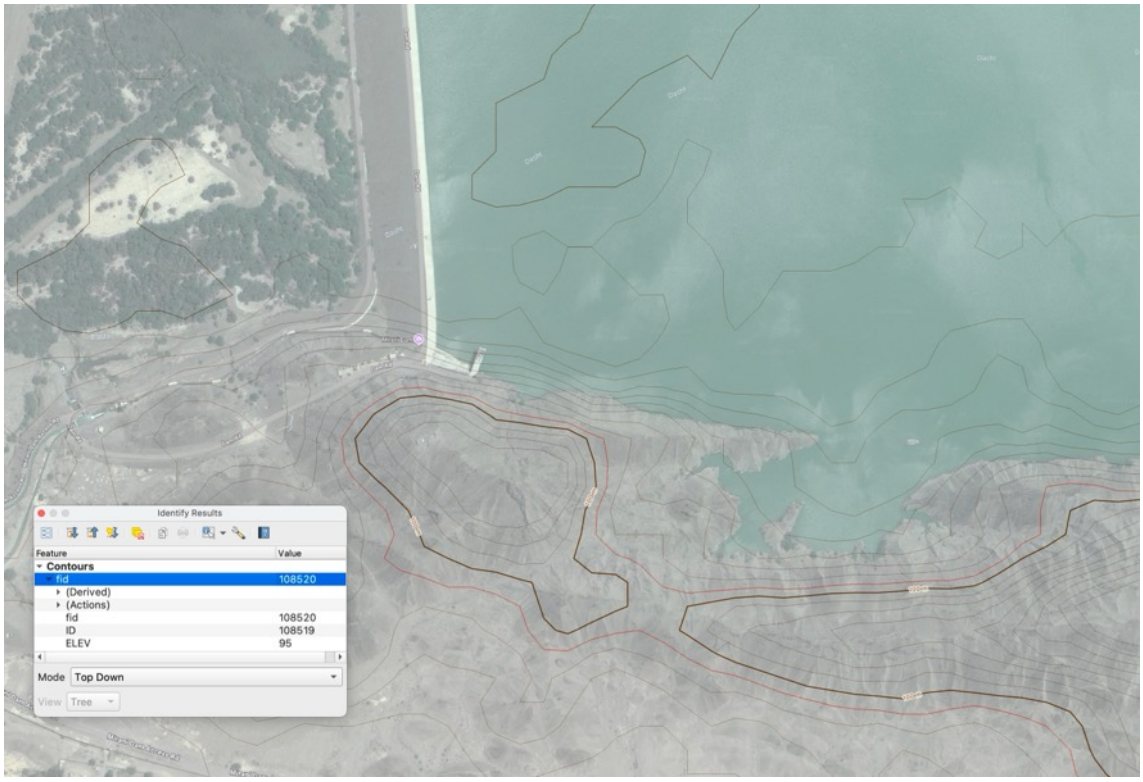
If the dam was built after February 2000, the contours represent the valley before the dam was built. To delineate the maximum area likely to be flooded by the reservoir, we can select a contour line which is higher than the edge of the dam where it meets the surrounding hillside.

- Put the contours layer on top of the background satellite layer
- Style the Contours: **Right click | Properties | Symbology | Layer Rendering | Style | Load Style**
- Select **1_Maps | 3_QML | Contours_SRTM30m_10m.qml**
- Press **Load style**
- Press **OK**

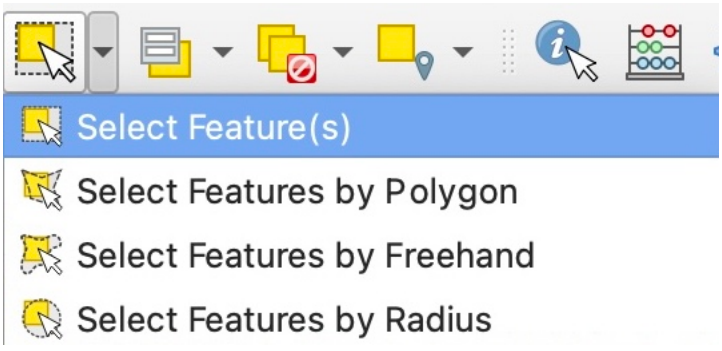




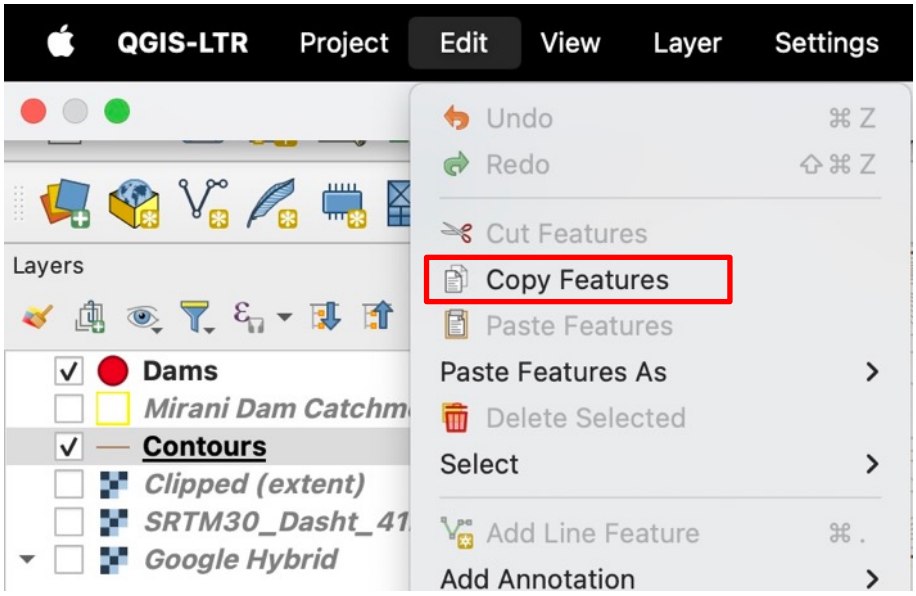
- Click on **Identify Feature(s)**
- If you cannot find it, make sure that the **Selection Toolbar** is activated
- Check the elevations of the contour lines which are above the current edge of the dam. Remember that these contours are based on the SRTM30, which was acquired before the dam was built, so they show the elevations before the ground water disturbed by the construction works.
- See how the selected contour line follows the outline of the water level.
- Select a contour which is at least 10-20m above the top of the dam. For the Mirani dam the 95m SRTM elevation is a good choice.

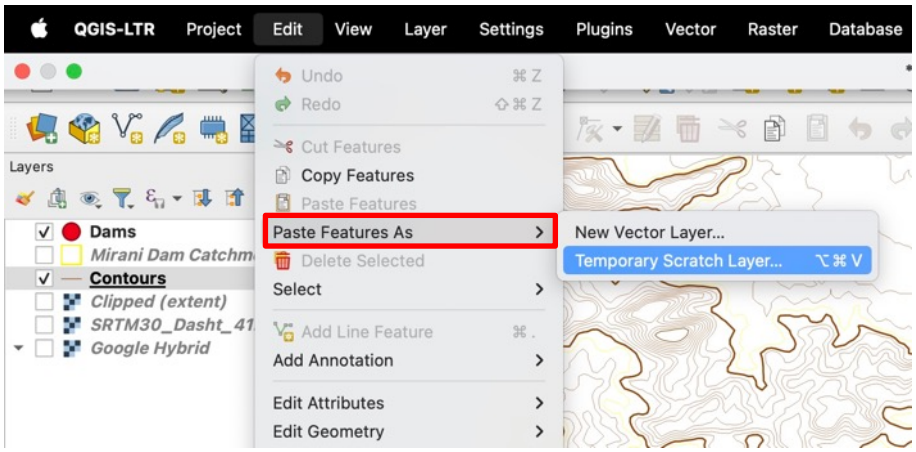


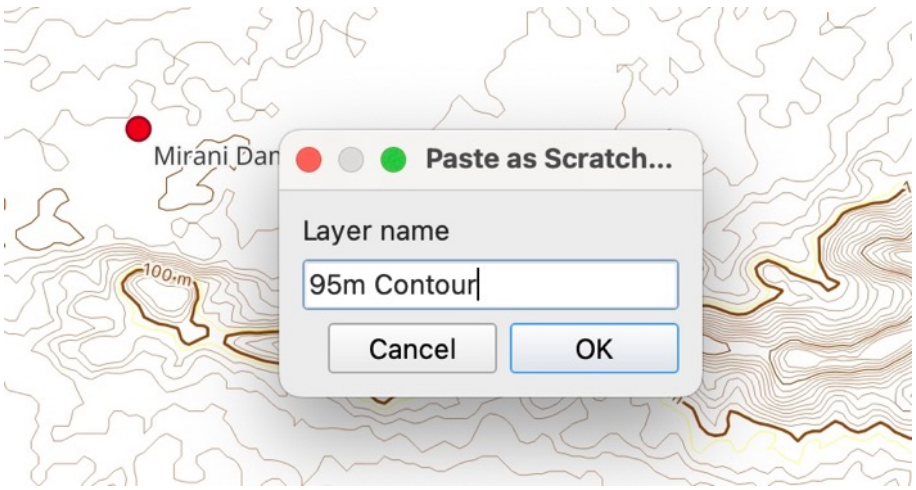
- Select the contours layer
- Click on **Identify Feature(s)**, which is part of the Selection Toolbar



- **Select the contour** which you have identified earlier. (95m for Mirani dam)
- On the tool bar at the top of the screen, click on **Edit | Copy Features**
- On the tool bar at the top of the screen, click on **Edit | Paste Features As | Temporary Scratch Layer…**
- Name the temporary layer using the DEM contour elevation. e.g., 95m Contour







- Remove unnecessary temporary layers: Contours

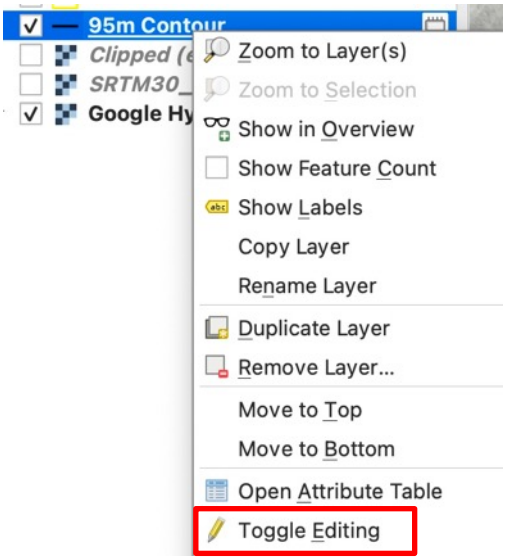
3.5 Edit the maximum flooded area contour

For the next step we need to zoom in on the dam, and have the following layers active:

- Temporary 95m contour
- Google Hybrid

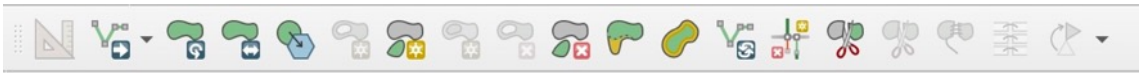
Now we will edit the contour line to delineate the maximum flooded area

- Select the temporary 95m contour
- Switch on editing mode: **Right-click | Toggle Editing** or click on the **yellow pencil** in the Digitising Toolbar



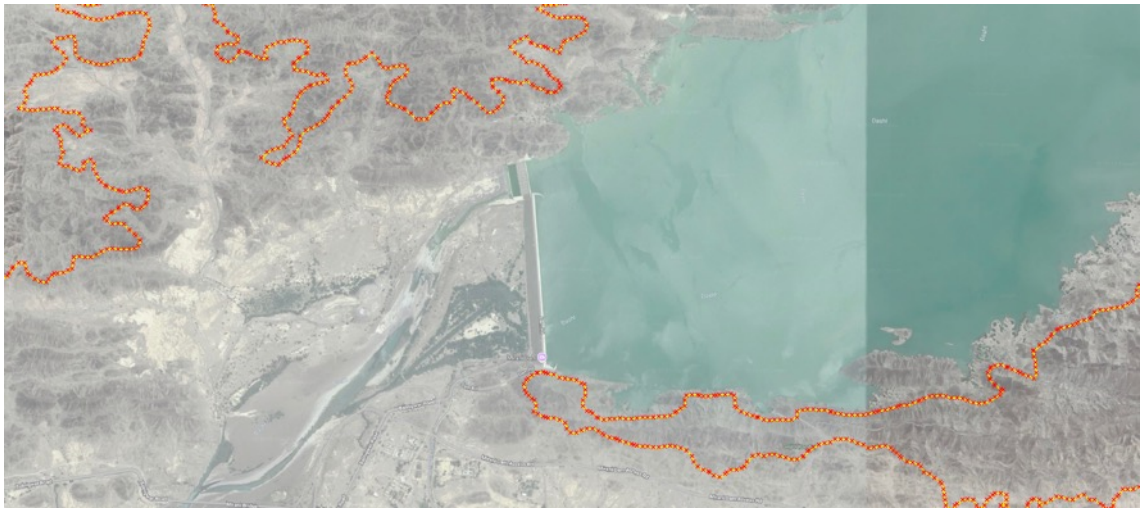
- Click on **View | Toolbars | Advanced Digitizing Toolbar** or
- Right-click in the top menu & select the **Advanced Digitizing Toolbar**

Note that the toolbar buttons are greyed out and only become activated if you have a layer in editing mode.

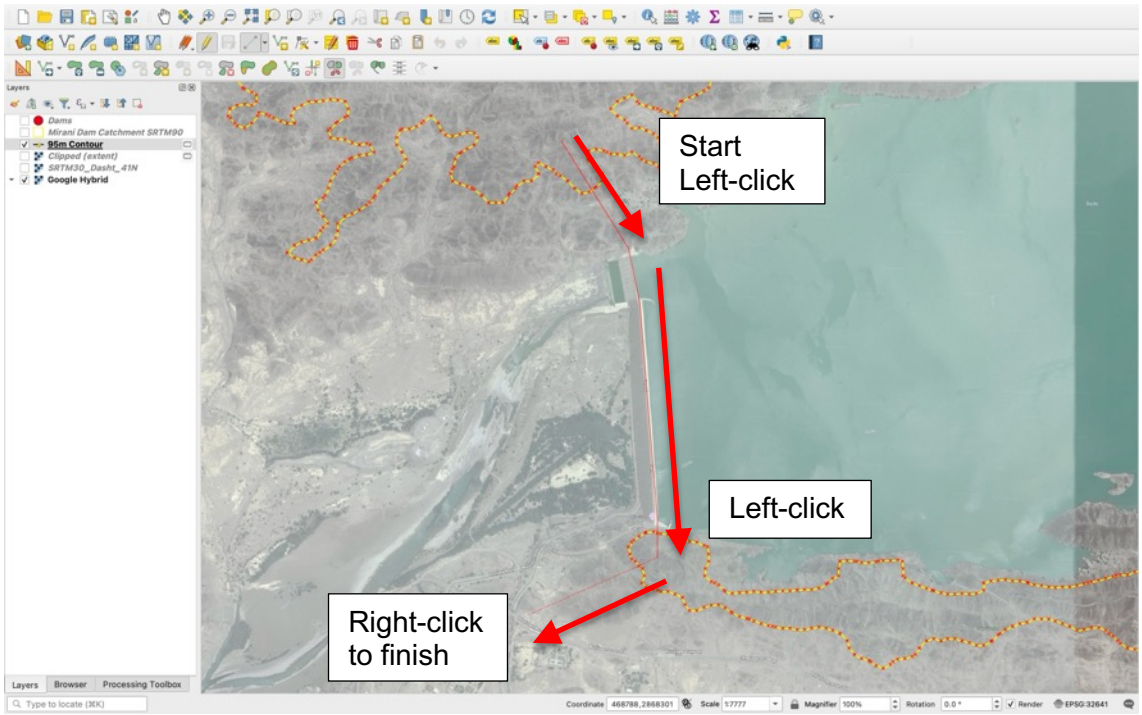


- **Select** the temporary **95m Contour** using the select button. 

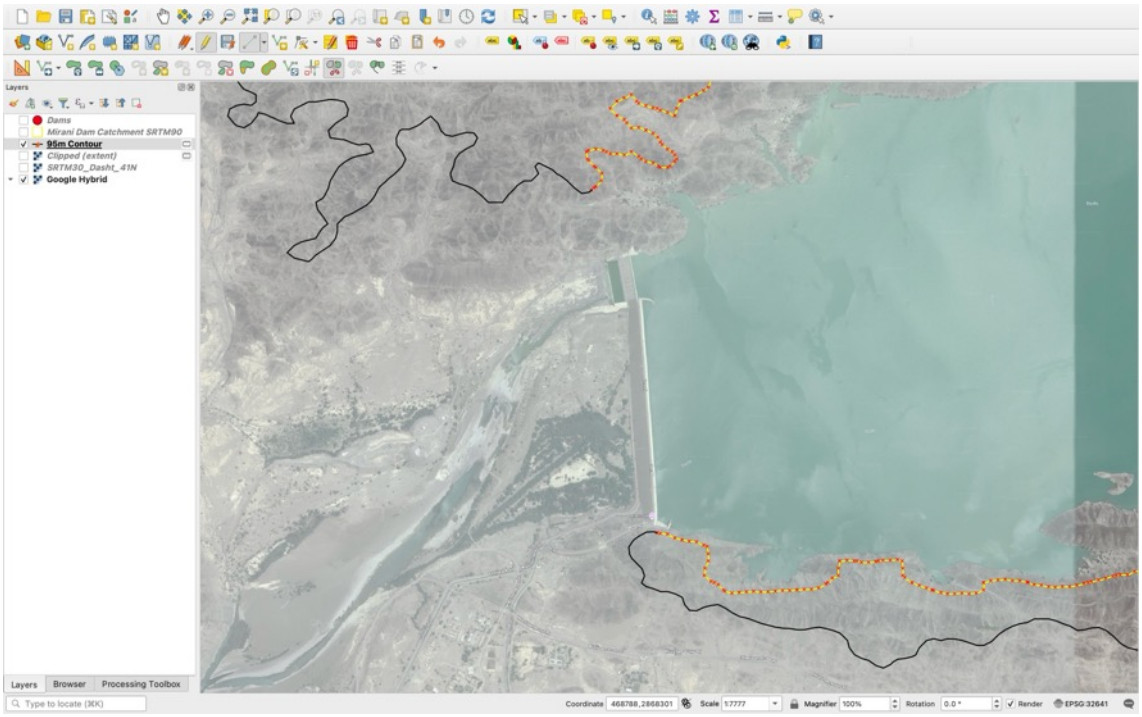
- Note that it turns yellow & you can see all the nodes as red crosses. It means that it now can be edited.



- In the *Advanced Editing Toolbar*, select the **Split Features** tool 
- Draw a line crossing the contour line where you want to cut it. The line is shown in red and can consist of two or more sections. When you finish drawing it, right-click, to split the contour line.

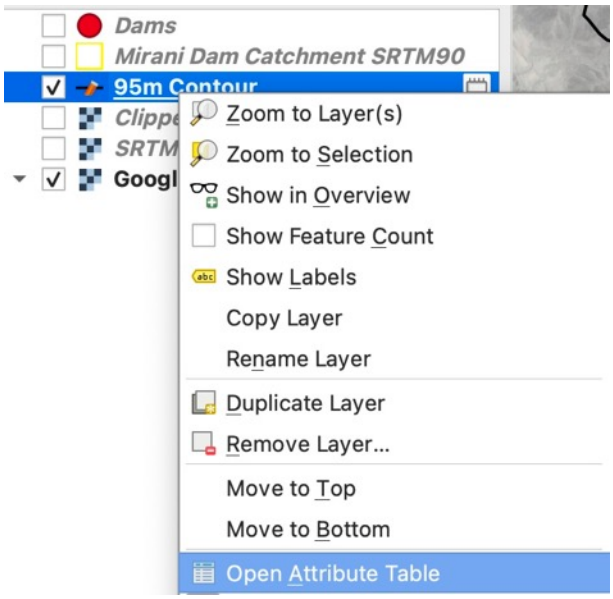


- If you have cut the contour line correctly you will see that one side is selected showing yellow colour with red crosses, while the other side is not. This will depend on which direction you did the cutting (clockwise or counterclockwise).

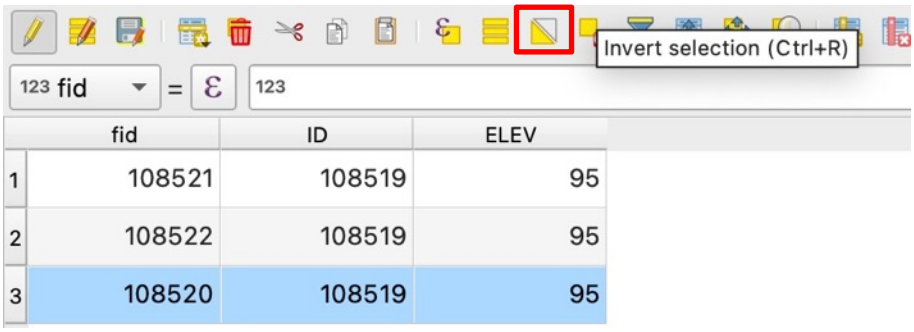


Now we will delete the contour lines which are downstream of the dam.

- Select the temporary 95m Contour & **Right-click | Open Attribute Table**

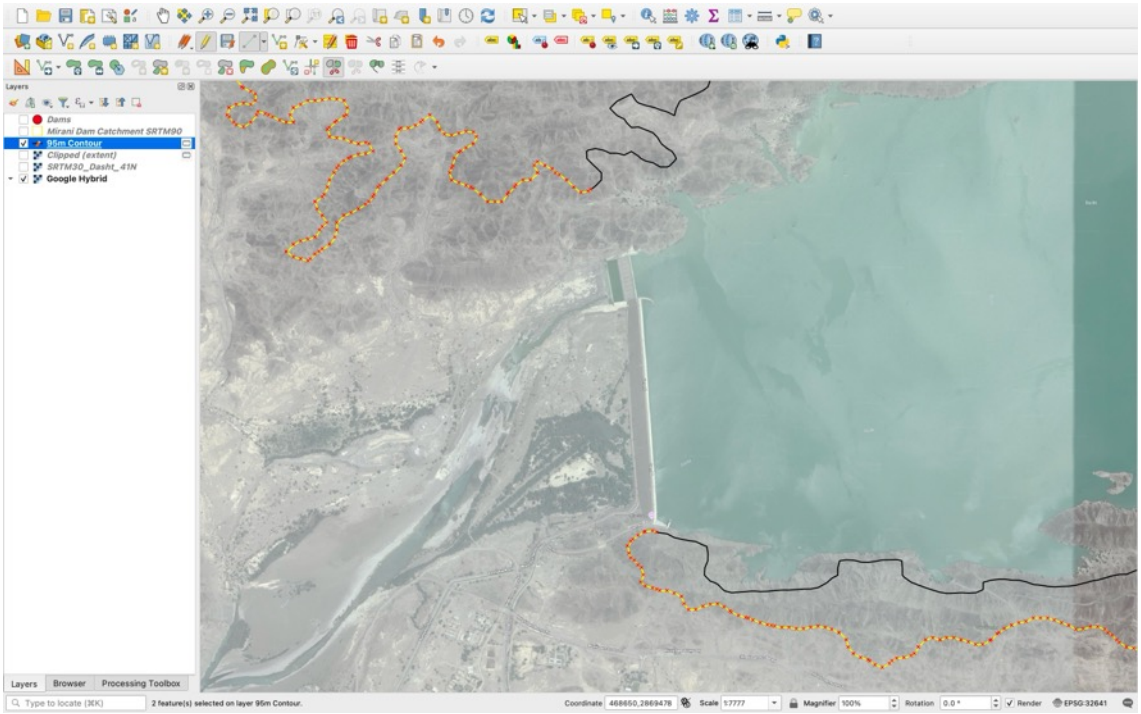


- If the contour line segment which has been selected is the one we want to keep, we need to Invert the selection (CTRL+R).

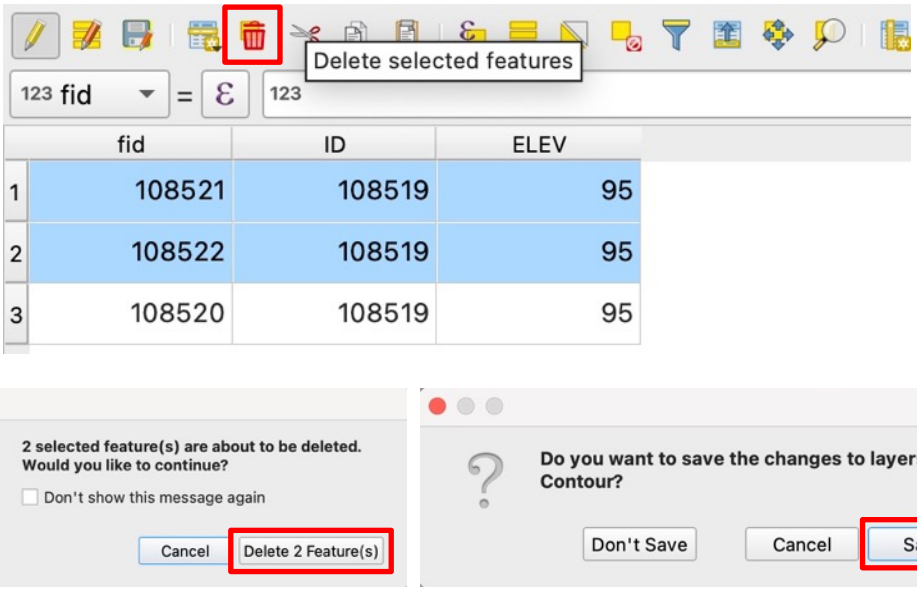


	fid	ID	ELEV
1	108521	108519	95
2	108522	108519	95
3	108520	108519	95

- Do a quick check on the Canvas to see that the contours you want to delete are selected.



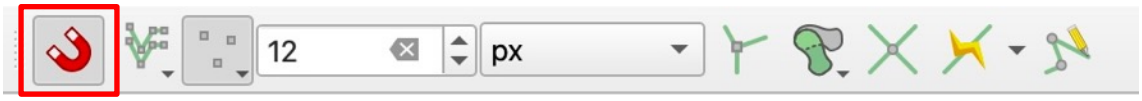
- In the Attribute Table, **Delete selected features**
- **Confirm** that you wish to delete them
- Click on the yellow pencil to **stop editing**
- Confirm that you want to **save the edits**



3.6 Digitise the dam axis & merge with the maximum flooded area contour

The next step is to add the dam axis and merge it with the contour, so we can delineate the total area upstream of the dam which can potentially be flooded.

- Right-click the top menu & activate the **Snapping Toolbar**
- Click on the red magnet. Note that it is set to snap to vertex within 12 pixels distance, which is usually OK.

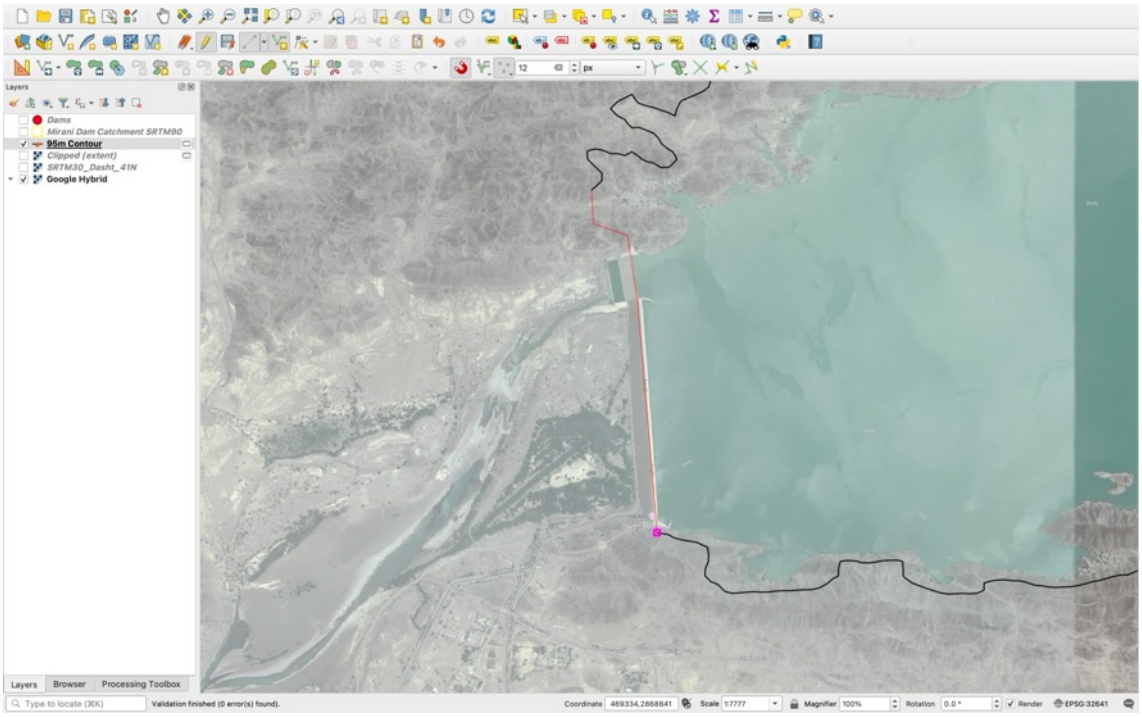


- Select the 95m Contour & toggle on editing using the **yellow pencil**

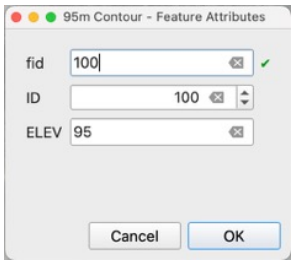
- In the **Digitizing Toolbar** click on Add Line Feature. 
- Move close to the existing contour line. When you get near a vertex, a **pink square** will appear. Click on it to start a new line exactly from an existing vertex.

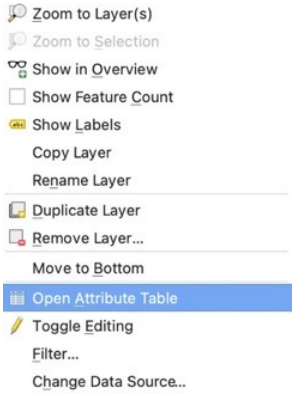


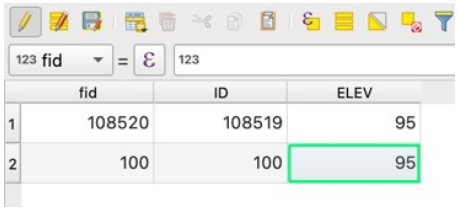
- Draw a line along the dam axis & along any ridges until you join up with the other end of the contour line. You can switch on the contour line layer to guide you.



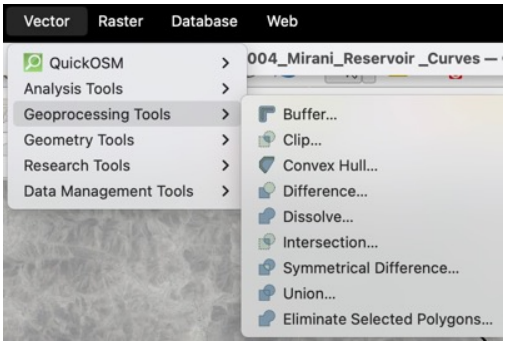
- You will get a pop-up. Add fid, ID and ELEV & Press OK
- Open the attribute table to **confirm that you have two polylines in the layer, and that both have the same elevation** i.e., ELEV = 95
- Click on the yellow pencil to stop editing & save your edits.

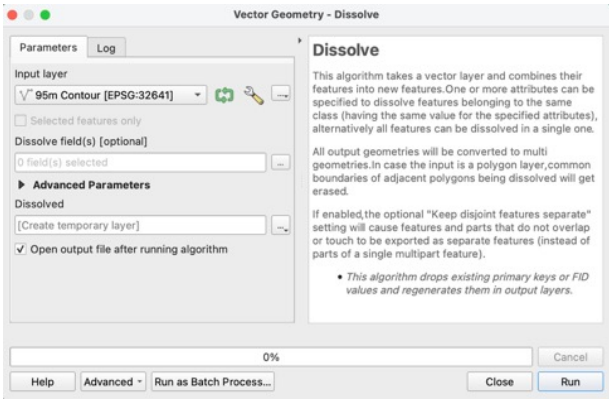


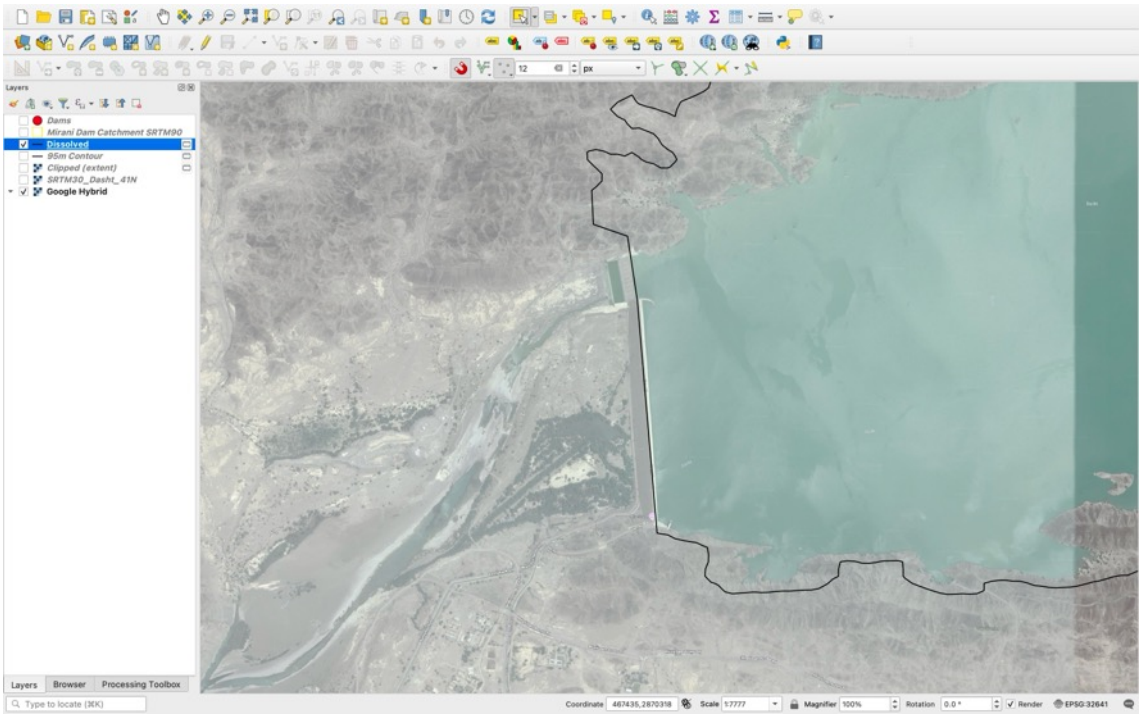




- Select **Vector | Geoprocessing Tools | Dissolve…**
- Make sure that the *95m Contour* layer is selected
- Optional: Choose **ELEV** for the Dissolve field.
- Press **Run**.
- A new temporary layer call *Dissolved* will be created.







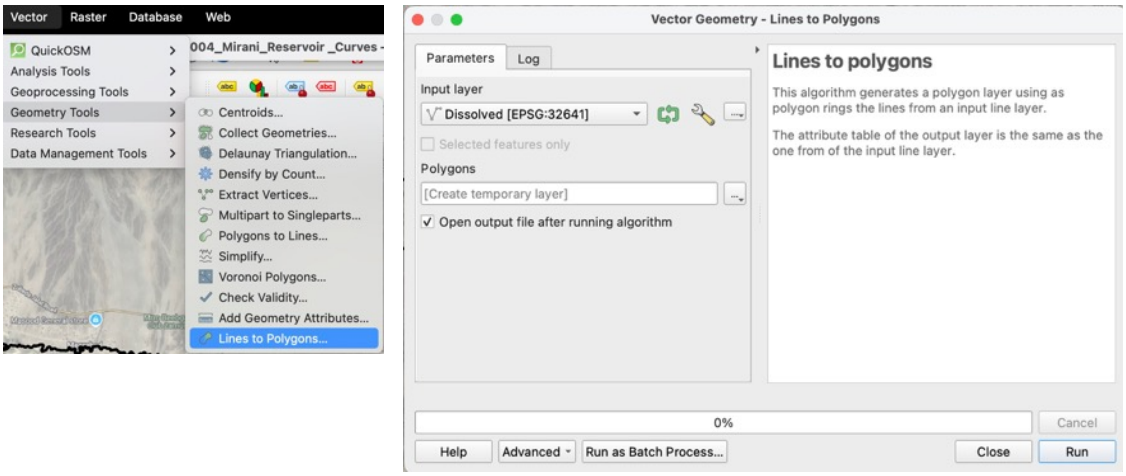


3.7 Convert maximum flooded area polyline to polygon

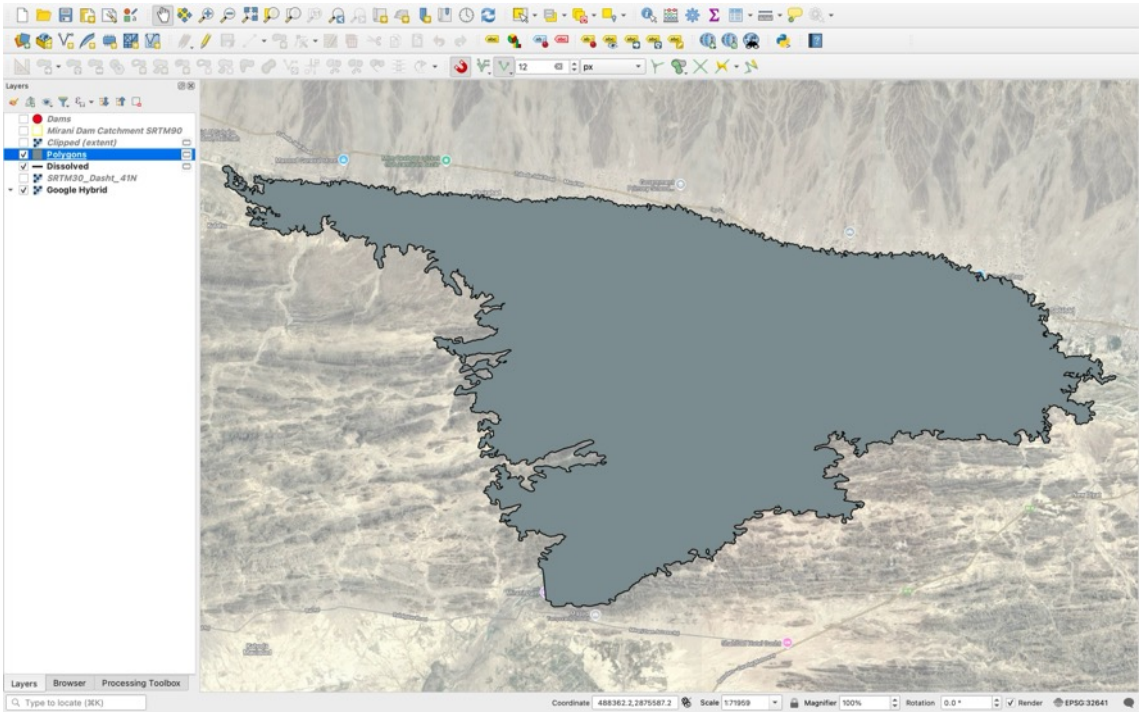
We now have a single polyline drawn from the dam axis to the maximum flooded area. This needs to be converted into a polygon which we will use to extract the digital elevation model and as a cookie-cutter for remote sensing data over the reservoir water body.

- Select the temporary *Dissolved* layer
- Select **Vector | Geometry Tools | Lines to Polygons**
- The Input layer should be the *Dissolved* layer.

- Press **Run**



- A temporary layer will be created called *Polygons*

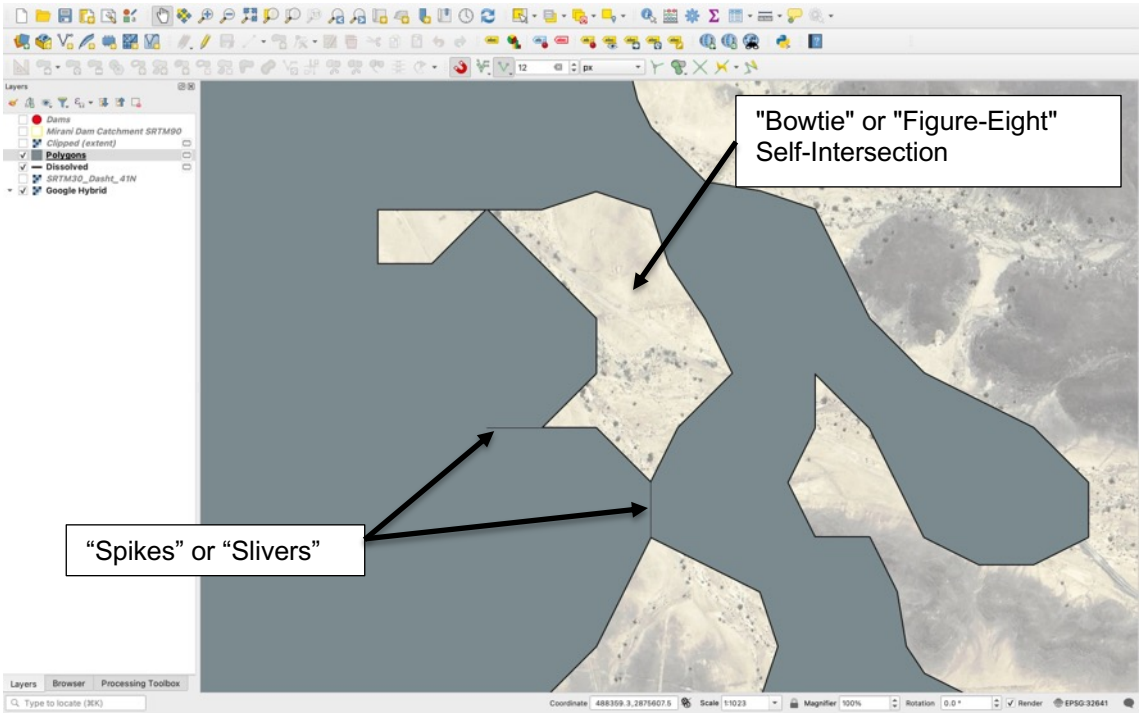


- Zoom into the periphery of the *Polygons* layer
- You may find two types of problems with the polygon related to **self-intersecting geometries**, which are related to the way in which the contour lines were derived from the digital elevation model. These are:

    - **Bowtie or Figure-Eight" Self-Intersections**. This occurs when a contour line, due to variations in terrain and the interpolation method, effectively folds back on itself and crosses its own path. Imagine a very narrow saddle or a sharp, isolated peak or depression. As the contour line attempts to encapsulate the terrain at a specific elevation, it might "pinch" and cross over itself. When these contour lines are later used to

form polygon boundaries (e.g., for an area between two contours representing an elevation band), this self-intersection directly translates into an invalid polygon shape. This is the most common and visually recognizable form of self-intersection.

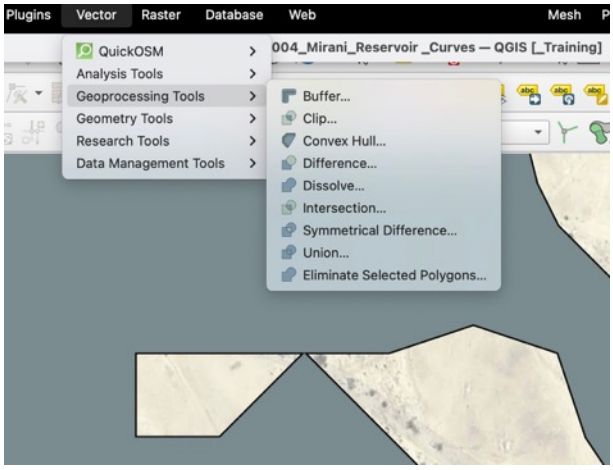
- **Interpolation Artifacts:** In some cases, the interpolation method used to generate contours from the DEM can create tiny, nearly zero-width “slivers” or "spikes" where a contour line briefly deviates and then immediately returns, causing two segments to lie almost perfectly on top of each other. These aren't distinct crossings but violate the topological rules of a simple polygon, leading to self-intersection errors in subsequent polygon creation or analysis.



These types of issues are highly problematic for GIS analysis as they violate the rules of simple, valid geometries and will cause errors with most geoprocessing tools in QGIS or other GIS software.

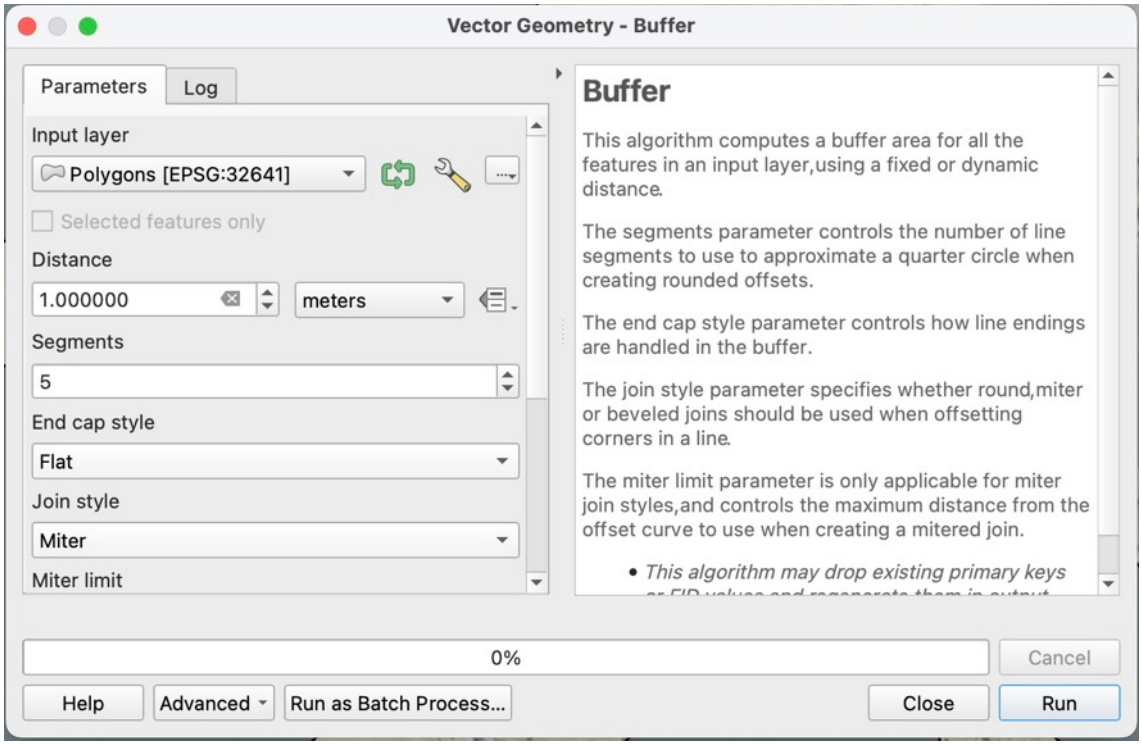
3.8 Fix self-intersecting geometries in the maximum flooded area polyline to polygon

- Select the Polygons layer
- Select **Vector | Geoprocessing Tools | Buffer**

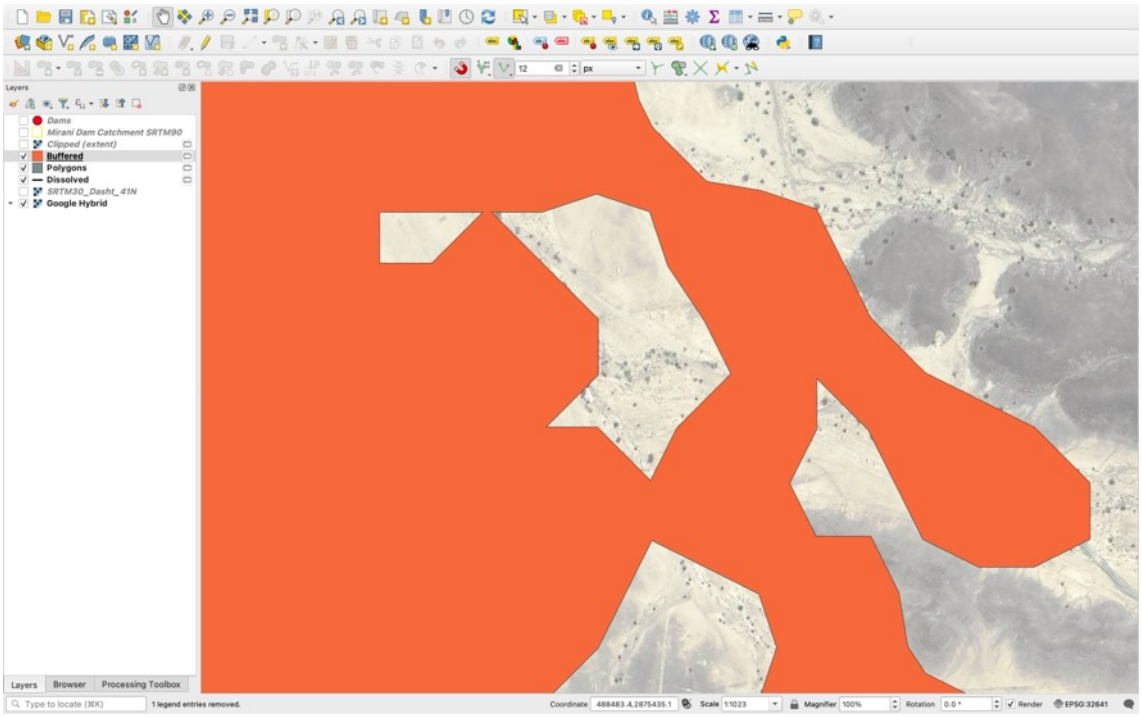


- **Input layer:** Polygons
- **Distance:** 1 metre
- **Segments:** 5
- **End cap style:** Flat
- **Join style:** Miter

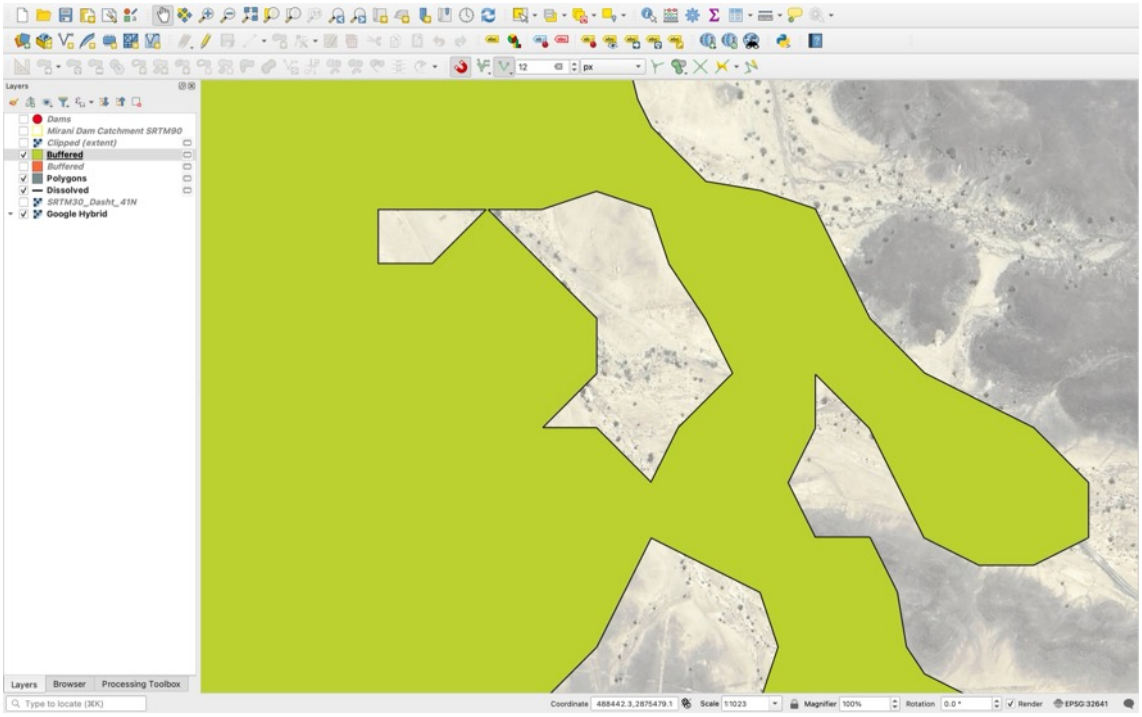
A new temporary layer will be created called Buffered



- **Check the shape of the Buffered layer:** Switch the Buffered layer on and off. It should be marginally larger than the Polygons layer but look the same. If it looks very different, go back one step & check the Buffer settings.
- **Note that this process has removed the Spike or Slivers.**

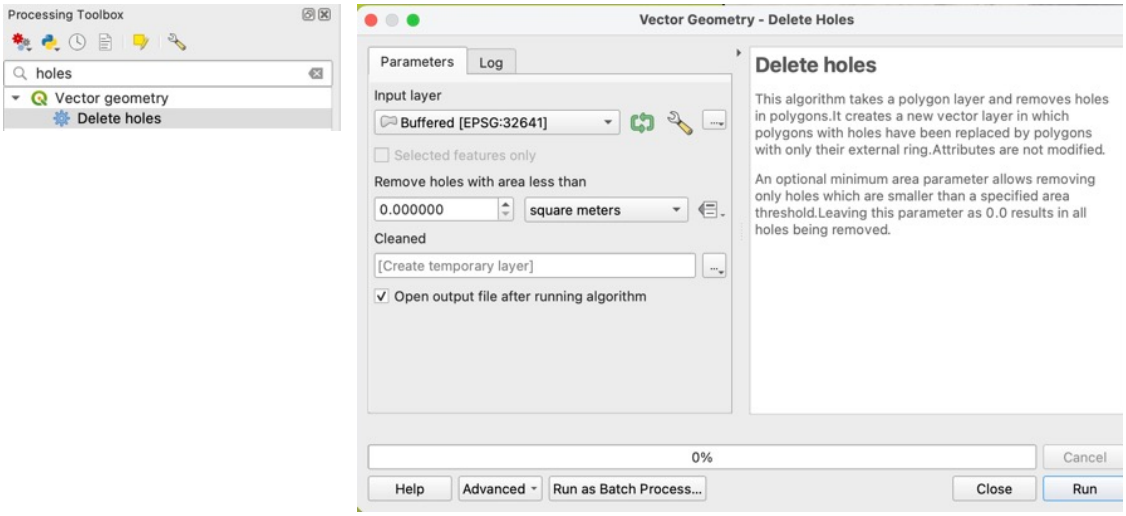


- Select the **Buffered** layer
- Select **Vector | Geoprocessing Tools | Buffer**
- Use the same settings as before, but **Distance:** -1 metre
- The polygon will be shrunk back to its correct size.
- Switch the new **Buffered layer** on and off. It should be exactly than same as the Polygons layer, except that there are no **Spike or Slivers.**

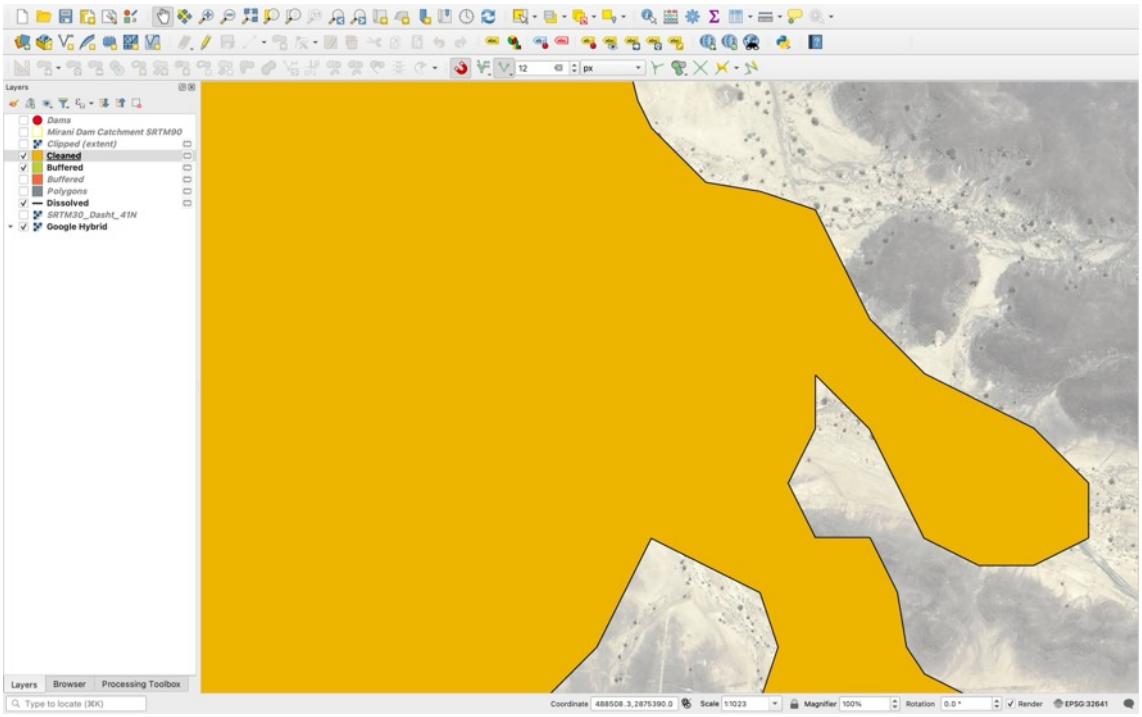


The only remaining problem is that the polygon contains holes, which need to be removed.

- Select the new **Buffered** layer
- In Processing Toolbox search for holes & select **Vector geometry | Delete holes**
- Leave all settings as they are & press **Run**

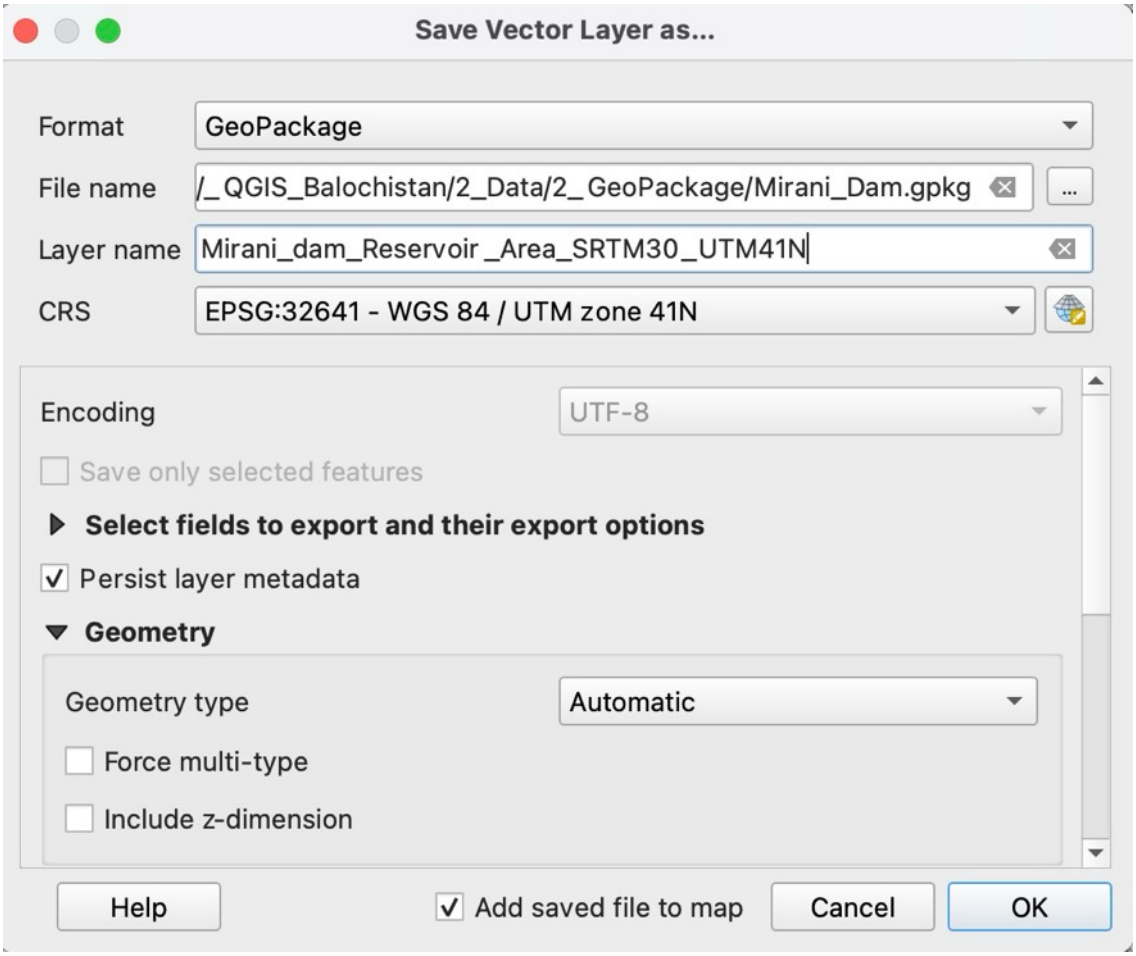


- A new layer called **Cleaned** has been created, free of all holes



- Select the **Cleaned** layer
- **Right-click | Export | save Features As…**

- **Format:** GeoPackage
- **File name:** under 2_Data/2_GeoPackage/Mirani_Dam.gpkg
- **Layer name:** Mirani_Dam_Reservoir_Area_SRTM30_UTM41N
- **CRS:** EPSG:432641 – WGS 84 / UTM zone 41N
- Press **OK**

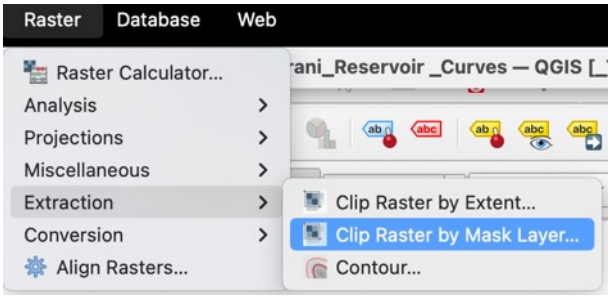


- Save the polygon again in geographic coordinates
- **Format:** GeoPackage
- **File name:** under 2_Data/2_GeoPackage/Mirani_Dam.gpkg
- **Layer name:** Mirani_Dam_Reservoir_Area_SRTM30_WGS84
- **CRS:** EPSG:4326 – WGS 84
- Press **OK**
- **Remove all the temporary polygon layers:** Cleaned, Buffered, Buffered, Polygons

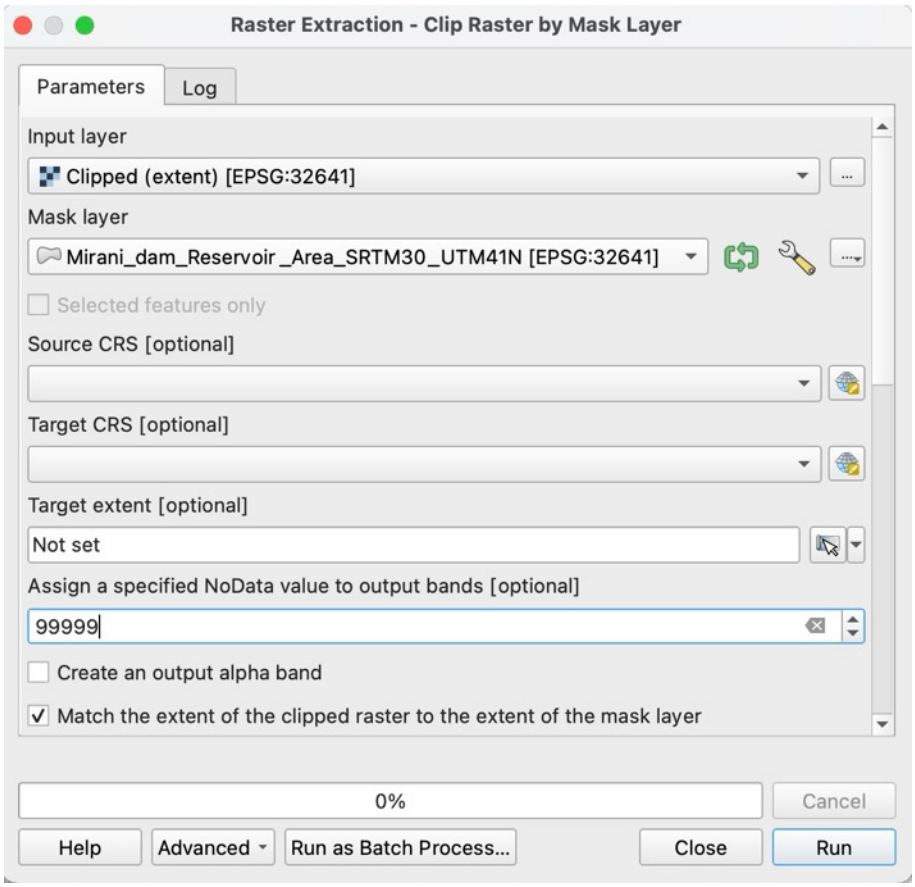
3.9 Create DEM of Reservoir flooded area

Now we will use the polygon of the maximum reservoir area to clip the digital elevation model.

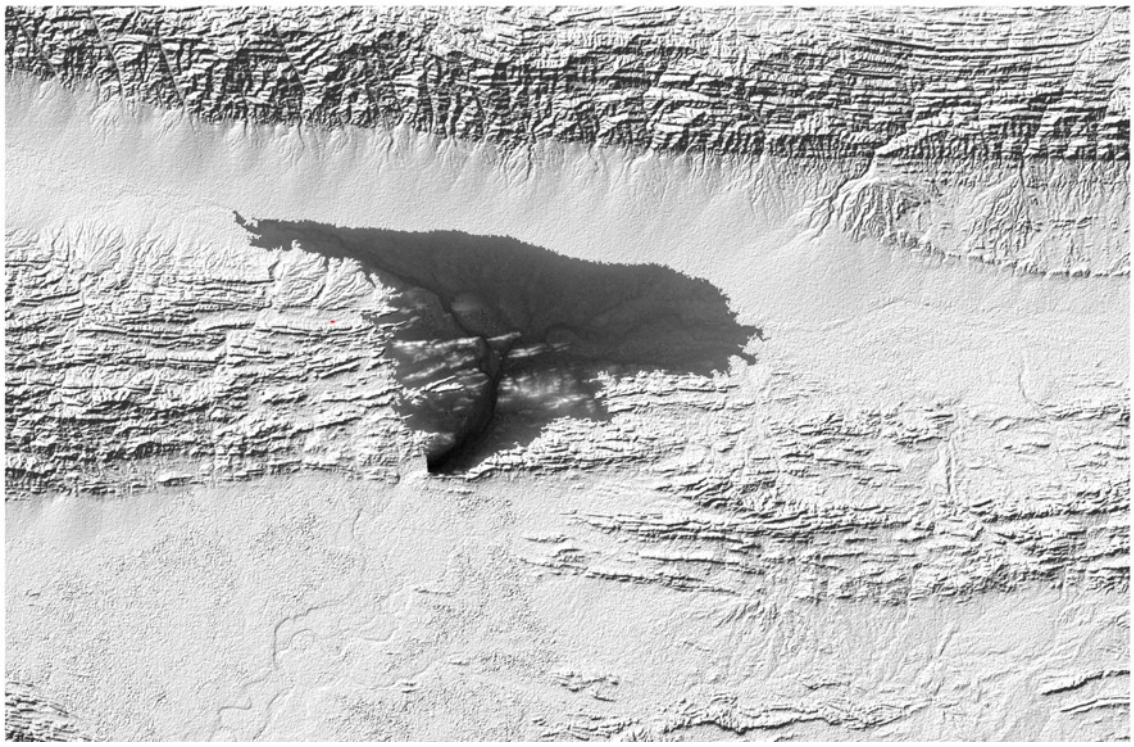
- Select the temporary **Clipped (extent)** digital elevation model.
- Select **Raster | Extraction | Clip Raster by Mask Layer…**



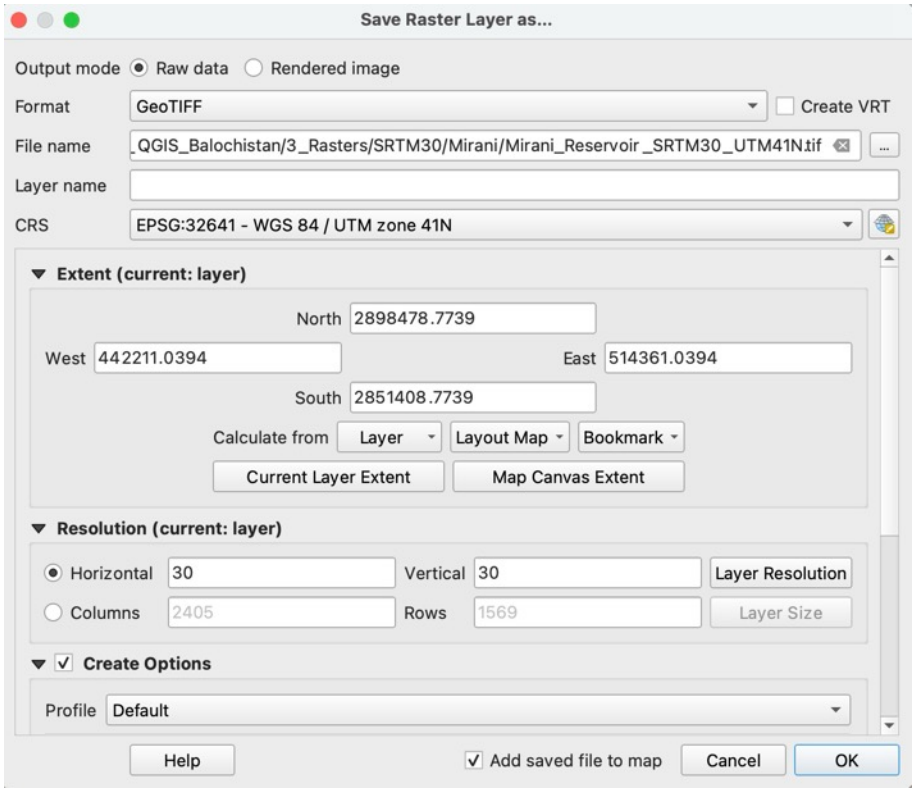
- **Input layer:** Clipped (extent)
- **Mask layer:** the projected reservoir polygon, Mirani_Dam_Reservoir_Area_SRTM30_UTM41N
- **Assign a specified no data value to output bands:** 99999
  For this use a value that does not occur in the DEM, such as 99999. If you forget to do this, the DEM will have a zero (0) elevation outside of the reservoir area, which will interfere with later calculations.
- Press **Run**.



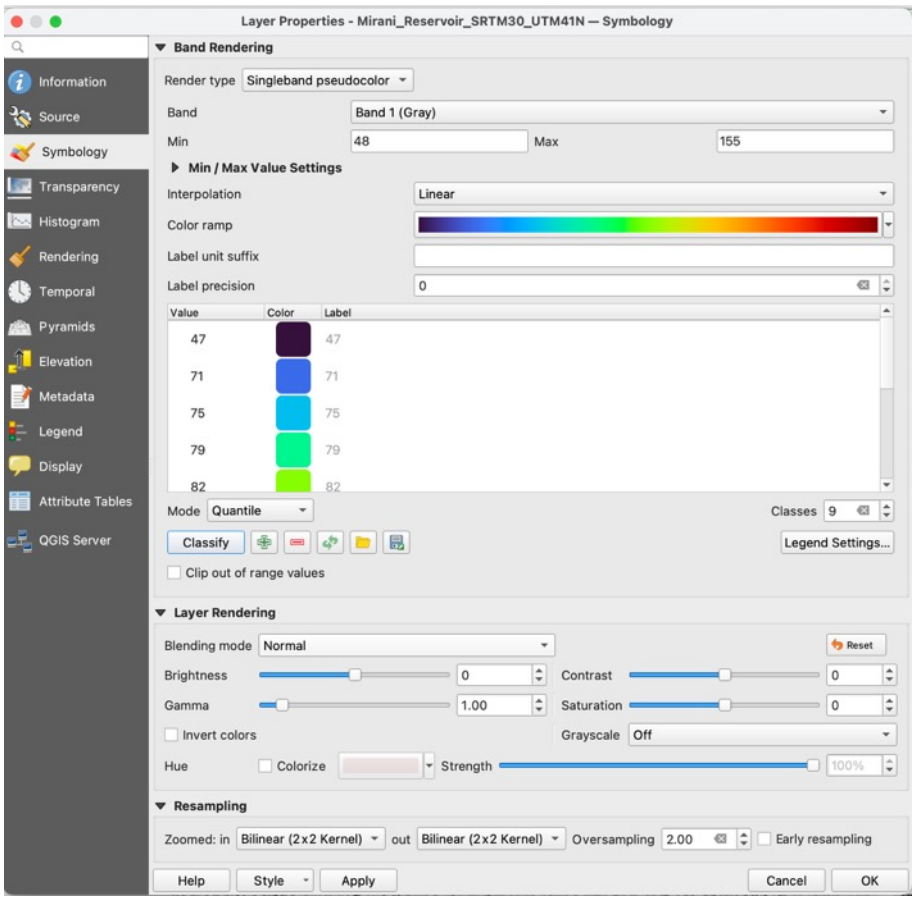
- A temporary raster layer has been produced called **Clipped (mask)**
- Check that the new DEM only covers the reservoir area, and that the elevation values outside of this have been set to **no data**.

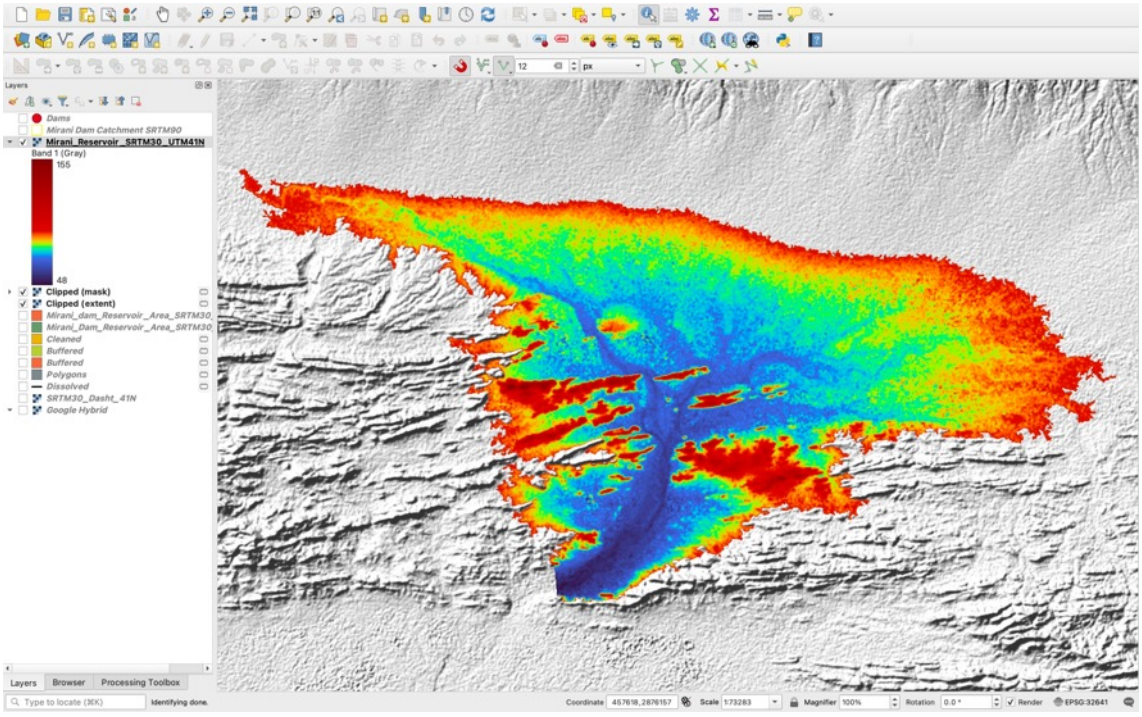


- Select **Clipped (mask)** & **Right-click | Export | Save As…**
- **Format:** GeoTIFF
- **File name:** 3_Rasters/SRTM30/Mirani/Mirani_Reservoir_SRTM30_UTM41N.tif
- **CRS:** EPSG:32641 – WGS 84 / UTM Zone 41
- Press **OK**



- **Remove all the temporary polygon layers:** Clipped (mask) & Clipped (extent)
- Set the reservoir DEM to a better colour scheme, such as Turbo

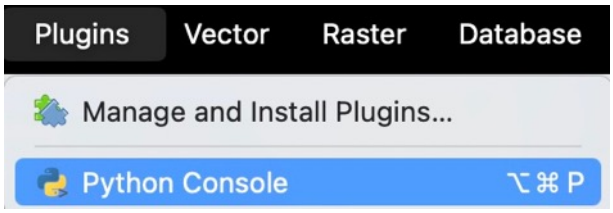


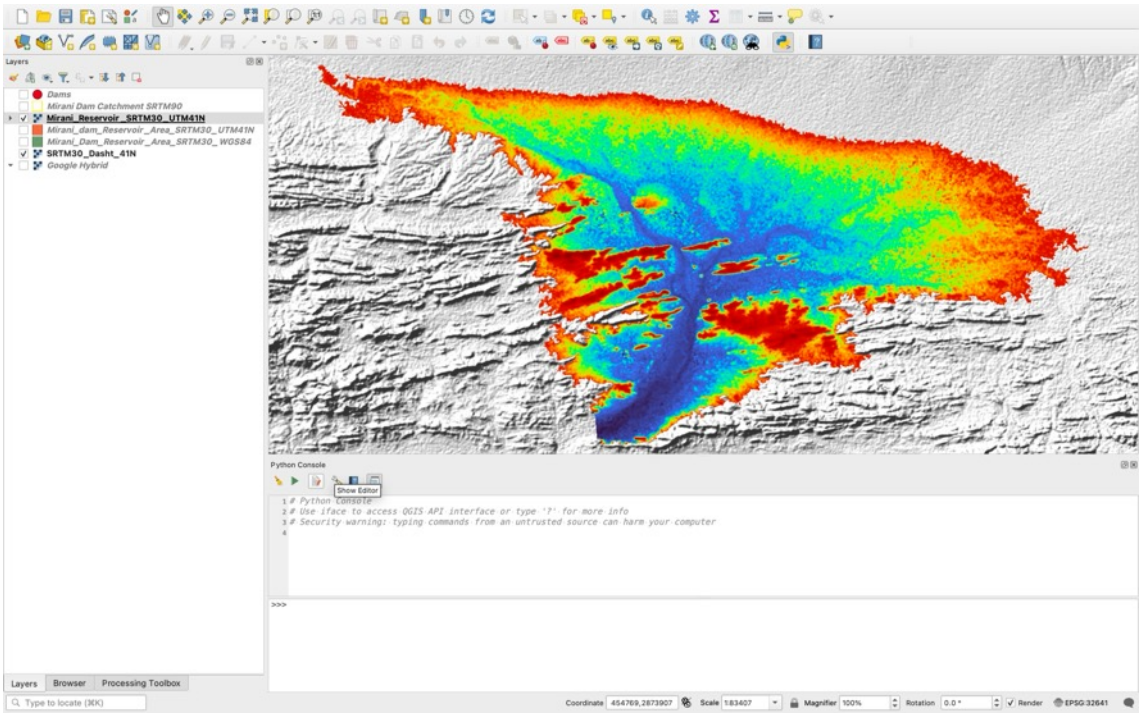


3.10 Calculate the Elevation – Area – Capacity curves

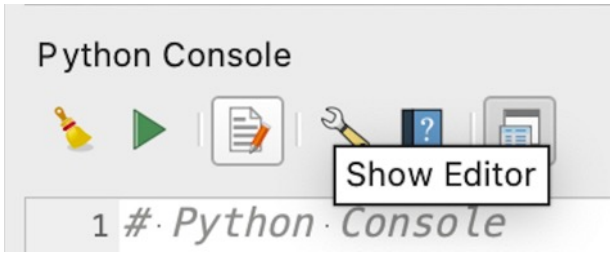
The projected DEM for the reservoir area is now correct & can be used for the calculation of the reservoir curves. Doing this in QGIS is quite complicated, so we will automate the process using a python script.

- **Select the DEM of the maximum flooded reservoir area:** *Mirani_Reservoir_SRTM30_UTM41N.tif*
- On the top menu select **Plugins | Python Console**
- Alternatively click on the **python icon** in the Plugins Toolbar

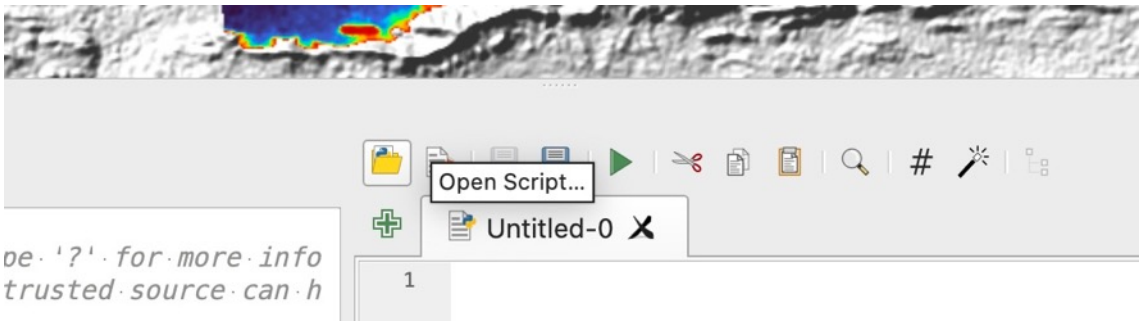




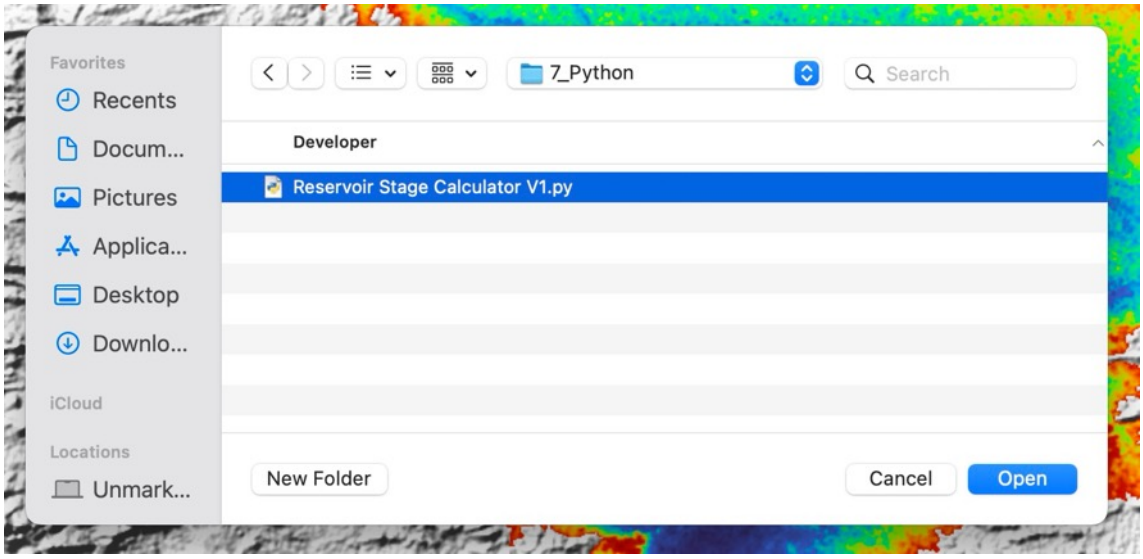
- In the Python Console, click on **Show Editor**.



- The Python Editor should open on the right.
- Click on the **Open Script…** icon

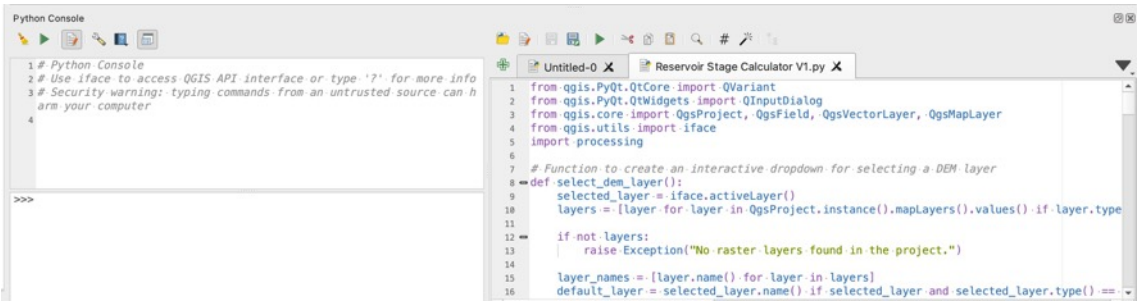


- Select the **Reservoir Stage Calculator V1.py** script from the *01_Maps/07_Python* folder.
- Press **Open**

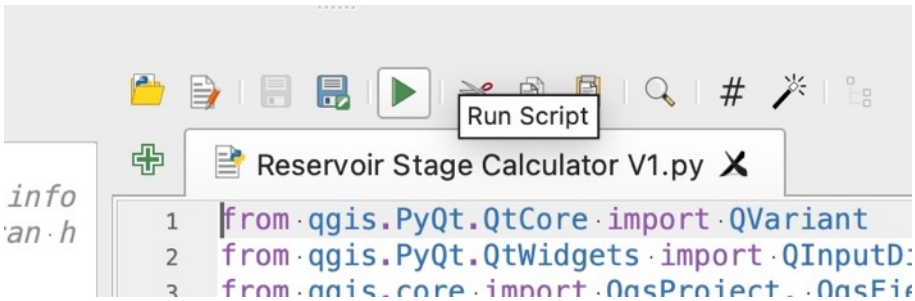


The script should load as shown below.

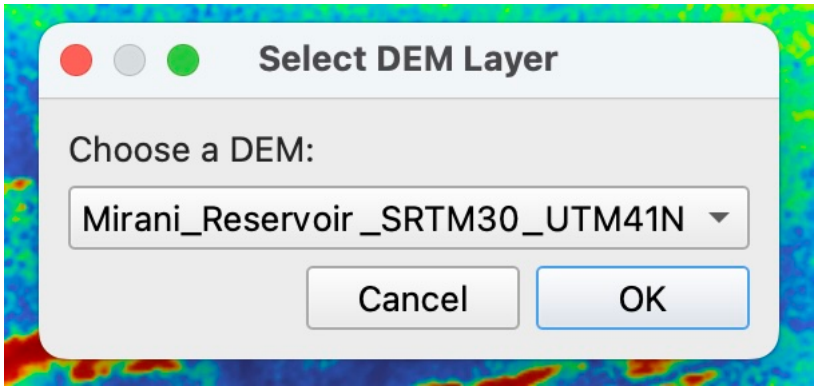
- Delete the **Untitled-0 script**



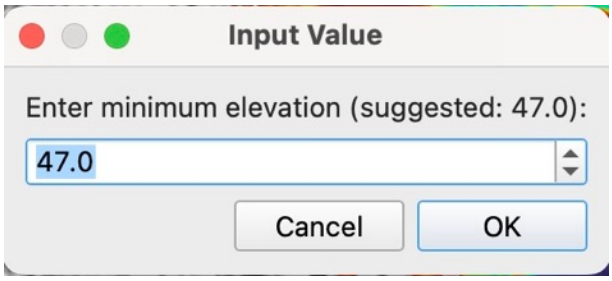
- Click on the green **Run Script** icon



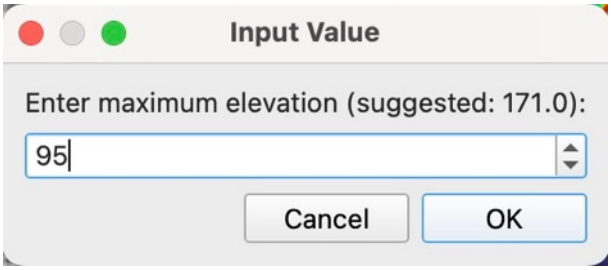
- **Choose a DEM:** If you have previously selected the reservoir DEM it will already be selected. If not, look at the available rasters in the drop-down list.
- Press **OK.**

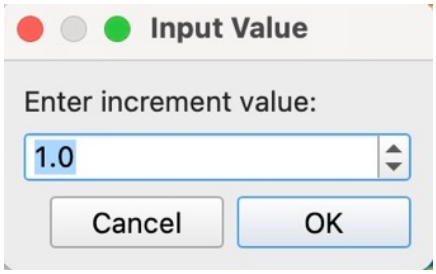


- **Enter minimum elevation:** The script will suggest the lowest value in the DEM. Usually this is the correct one to use.



- **Enter Maximum elevation:** The script will suggest the highest value in the DEM. For the Mirani reservoir it will pick the highest points of some of the islands in the reservoir. It is advisable to use the same elevation as what was used to delineate the DEM. i.e., 95m for the Mirani Reservoir
- Press **OK**
- **Enter increment value:** The script will suggest 1.0 metres, which is usually a good one to use.
- Press **OK**

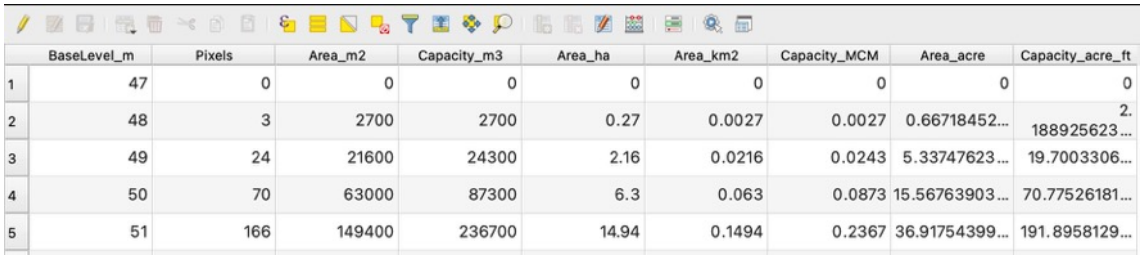




A new temporary table will be created which starts with *Elevation_Area_Volume_* followed by the name of the DEM.



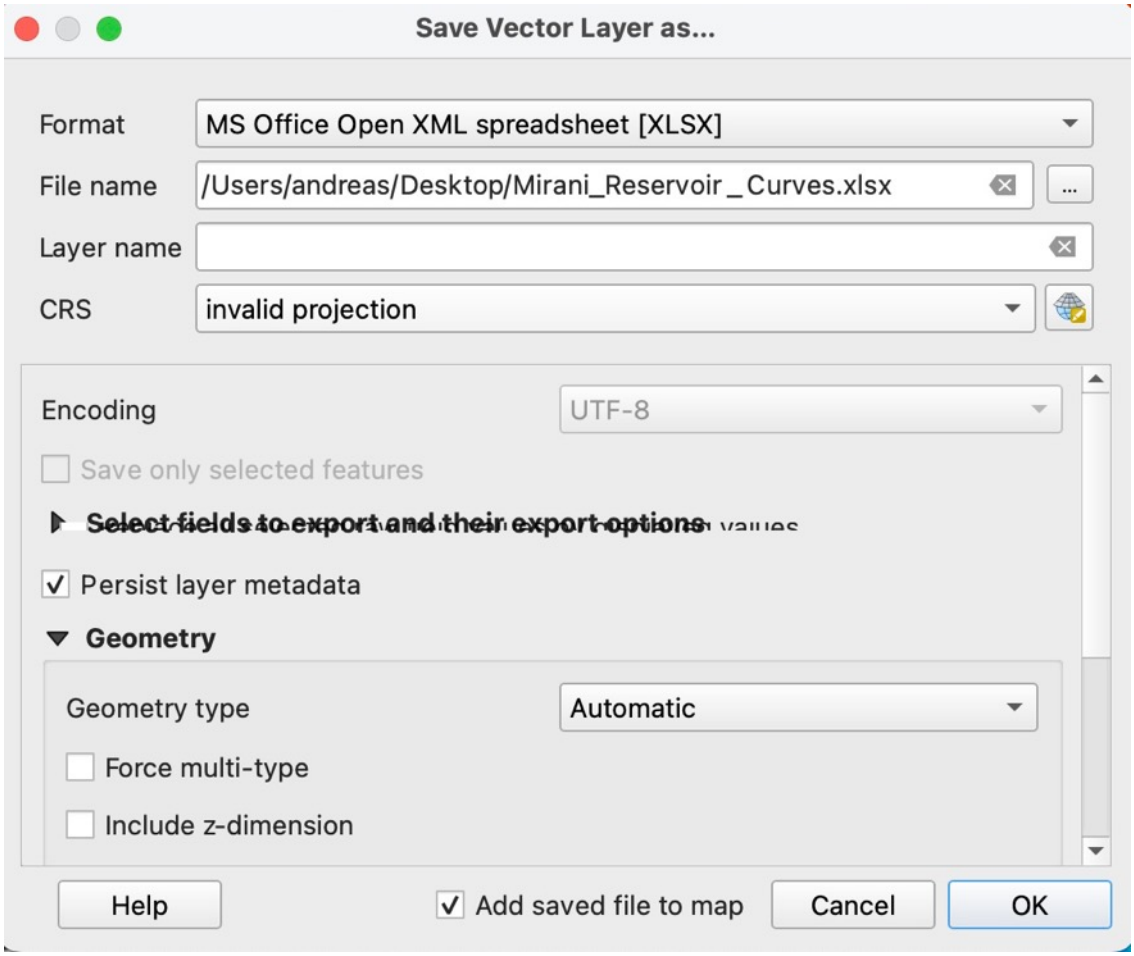
- Select the new table
- **Right-click | Open Attribute Table**



	BaseLevel_m	Pixels	Area_m2	Capacity_m3	Area_ha	Area_km2	Capacity_MCM	Area_acre	Capacity_acre_ft
1	47	0	0	0	0	0	0	0	0
2	48	3	2700	2700	0.27	0.0027	0.0027	0.66718452...	2.188925623...
3	49	24	21600	24300	2.16	0.0216	0.0243	5.33747623...	19.7003306...
4	50	70	63000	87300	6.3	0.063	0.0873	15.56763903...	70.77526181...
5	51	166	149400	236700	14.94	0.1494	0.2367	36.91754399...	191.8958129...

Note that for each elevation called BaseLevel_m, there are the number of Pixels, the Area_m2, Capacity_m3, Area_ha, Area_km2, Capacity_MCM, Area_acre and Capacity_acre_ft.

- Close the Attribute Table
- Select the *Elevation_Area_Volume_* table
- **Right-click | Export | Save Features As…**
- **Format:** MS Office Open XML spreadsheet [XLSX]
- **File name:** Mirani_Reservoir_Curves
- Press **OK**



All the reservoir curve data has been exported to MS Excel.

	A	B	C	D	E	F	G	H	I
1	BaseLevel_m	Pixels	Area_m2	Capacity_m3	Area_ha	Area_km2	Capacity_MCM	Area_acre	Capacity_acre_ft
2	47	0	0	0	0	0	0	0	0
3	48	3	2700	2700	0.27	0.0027	0.0027	0.66718453	2.188925623
4	49	24	21600	24300	2.16	0.0216	0.0243	5.33747624	19.70033061
5	50	70	63000	87300	6.3	0.063	0.0873	15.56763903	70.77526182
6	51	166	149400	236700	14.94	0.1494	0.2367	36.91754399	191.895813
7	52	331	297900	534600	29.79	0.2979	0.5346	73.61269314	433.4072734
8	53	594	534600	1069200	53.46	0.5346	1.0692	132.1025369	866.8145468

- **Delete the temporary table:** Elevation_Area_Volume_Mirani_Reservoir_SRTM30_UTM41N

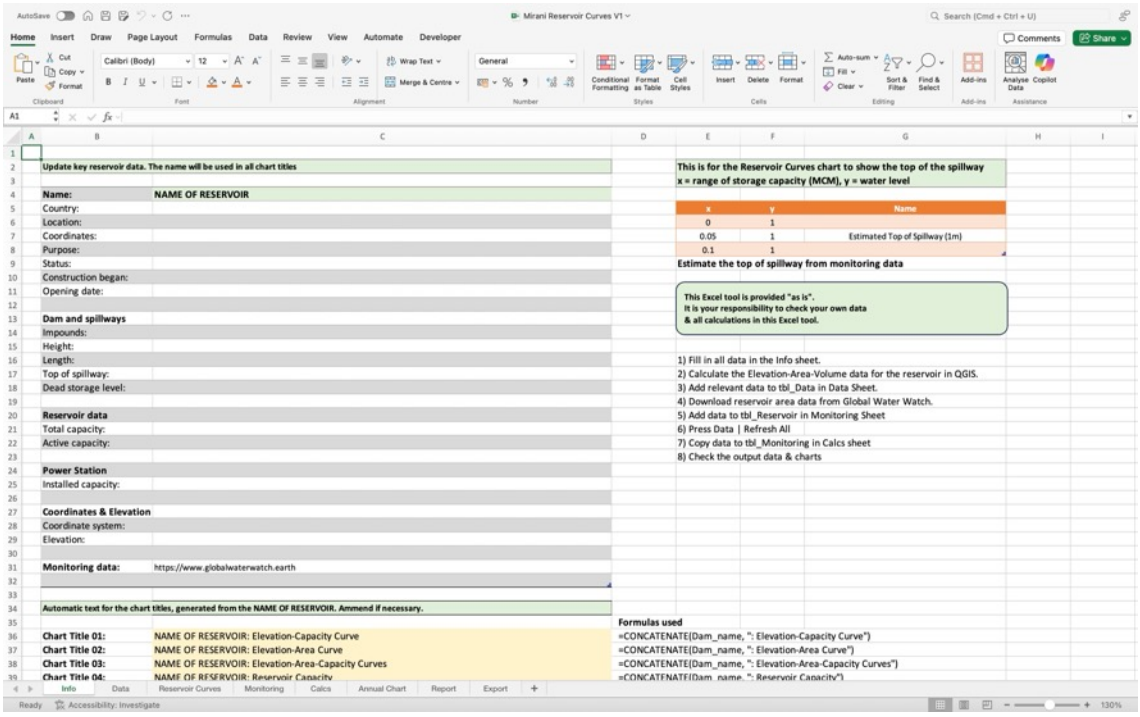


4 Process Reservoir Curves in MS Excel

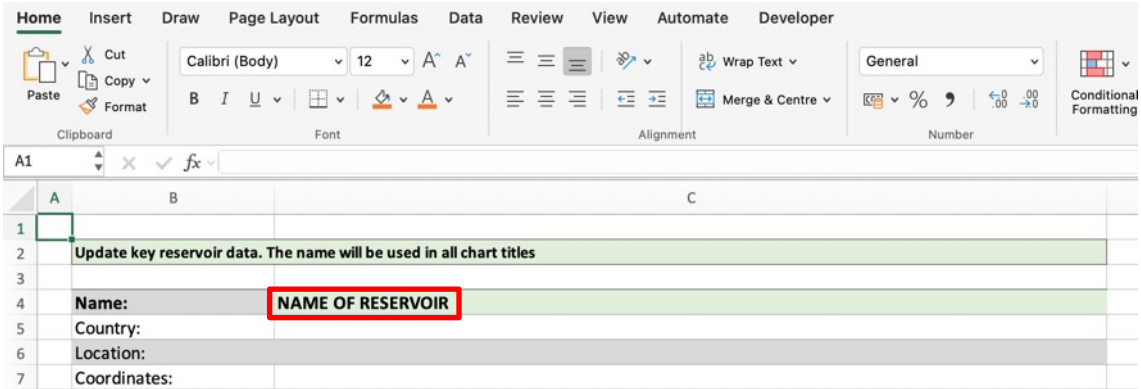
Once you have extracted the reservoir curve data in QGIS you can continue the processing in Excel. We will use a template to automatically draw the Elevation – Area – Capacity curves. The template can also process reservoir area monitoring data from remote sensing sources to calculate reservoir capacity and water levels.

4.1 Open Reservoir Curves Template in Excel

- Make a copy of the **Reservoir Curves Template.xlsx**
- Rename it to the name of your reservoir e.g., **Mirani Reservoir Curves.xlsx**
- **Open the template in Excel.** There should not be any security warnings as there is no active content.



- Replace the *NAME OF RESERVOIR* with **Mirani Reservoir, Balochistan**

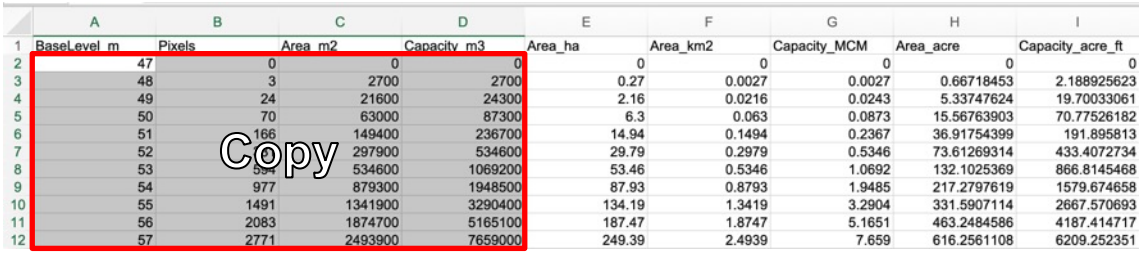


- Note that this name will be used to create the chart titles in the orange box below.
- Fill in the rest of the reservoir data if available. This is just for information but could be useful for the interpretation of the data.

> ***Note the disclaimer:***
> *This Excel tool is provided "as is". It is your responsibility to check your own data & all calculations in this Excel tool.*

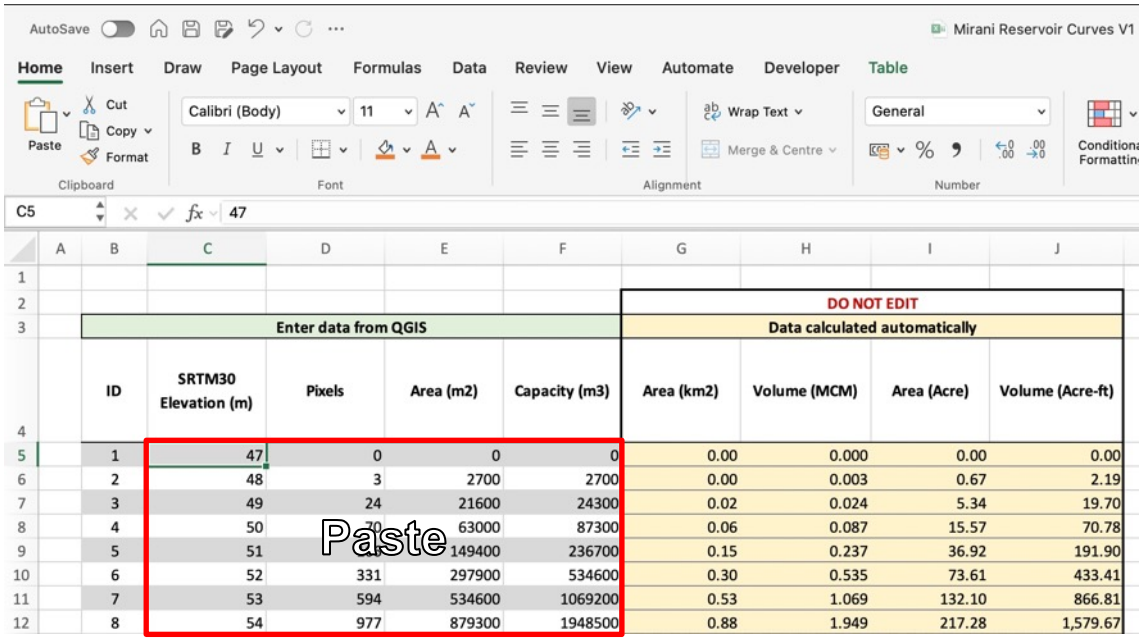
4.2 Copy-Paste the reservoir curves data

- Open the reservoir curves Excel workbook which you exported from QGIS, **Mirani_Reservoir_Curves.xlsx**
- **Copy the data from the first four (4) columns** in the sheet i.e., BaseLevel_m to Capacity_m3



	A	B	C	D	E	F	G	H	I
1	BaseLevel_m	Pixels	Area_m2	Capacity_m3	Area_ha	Area_km2	Capacity_MCM	Area_acre	Capacity_acre_ft
2	47	0	0	0	0	0	0	0	0
3	48	3	2700	2700	0.27	0.0027	0.0027	0.66718453	2.188925623
4	49	24	21600	24300	2.16	0.0216	0.0243	5.33747624	19.70033061
5	50	70	63000	87300	6.3	0.063	0.0873	15.56763903	70.77526182
6	51	166	149400	236700	14.94	0.1494	0.2367	36.91754399	191.895813
7	52	331	297900	534600	29.79	0.2979	0.5346	73.61269314	433.4072734
8	53	594	534600	1069200	53.46	0.5346	1.0692	132.1025369	866.8145468
9	54	977	879300	1948500	87.93	0.8793	1.9485	217.2797619	1579.674658
10	55	1491	1341900	3290400	134.19	1.3419	3.2904	331.5907114	2667.570693
11	56	2083	1874700	5165100	187.47	1.8747	5.1651	463.2484586	4187.414717
12	57	2771	2493900	7659000	249.39	2.4939	7.659	616.2561108	6209.252351

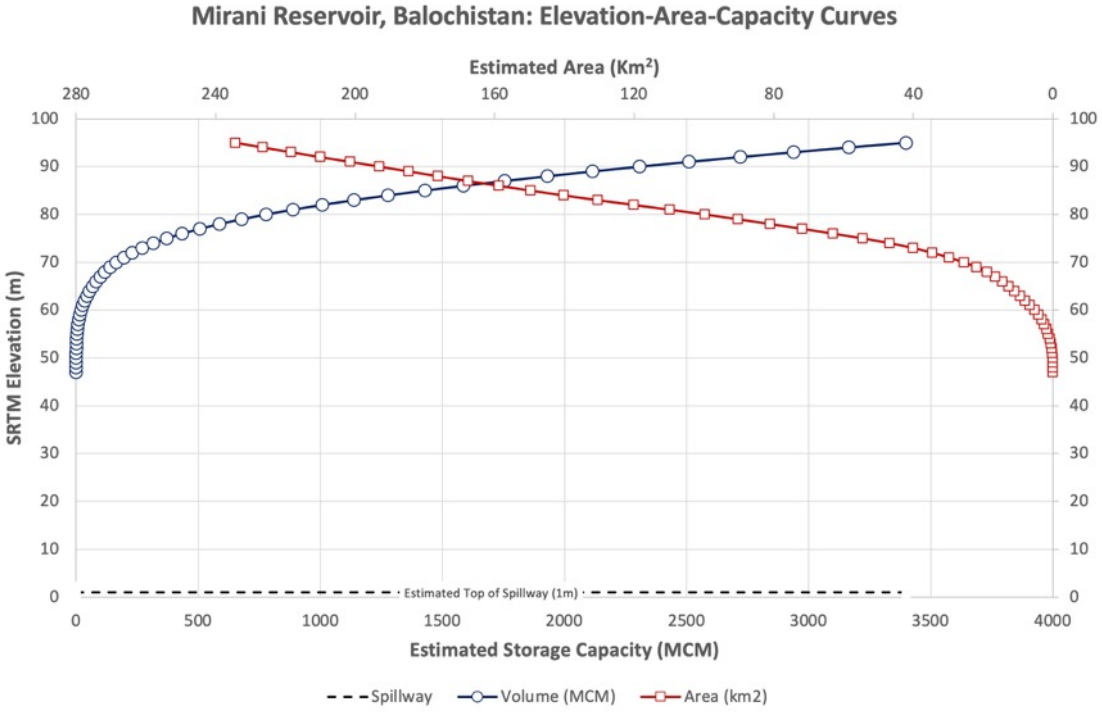
- Go to the **Mirani Reservoir Curves.xlsx | Data Sheet**
- **Paste the data from SRTM30 Elevation (m) to Capacity (m3)**



	ID	SRTM30 Elevation (m)	Pixels	Area (m2)	Capacity (m3)	Area (km2)	Volume (MCM)	Area (Acre)	Volume (Acre-ft)
5	1	47	0	0	0	0.00	0.000	0.00	0.00
6	2	48	3	2700	2700	0.00	0.003	0.67	2.19
7	3	49	24	21600	24300	0.02	0.024	5.34	19.70
8	4	50	70	63000	87300	0.06	0.087	15.57	70.78
9	5	51	166	149400	236700	0.15	0.237	36.92	191.90
10	6	52	331	297900	534600	0.30	0.535	73.61	433.41
11	7	53	594	534600	1069200	0.53	1.069	132.10	866.81
12	8	54	977	879300	1948500	0.88	1.949	217.28	1,579.67

4.3 Edit the Reservoir Curves presentation

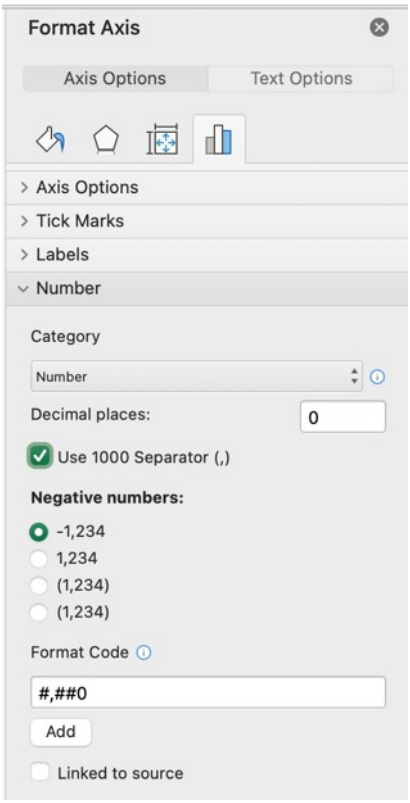
- Go to the **Reservoir Curves** Sheet
- This chart is linked to the Data sheet. **Check that the curves look OK**. They should be nice and smooth in big reservoirs like the Mirano Dam but can be "bumpier" for smaller reservoirs where the DEM pixel size is large compared with the reservoir area.
- Note the ***Estimated Top of Spillway (1m)*** dashed line along the bottom. This is **controlled from a table in the Info sheet**. Unless you have the exact elevation of the spillway and you know how to convert this from whatever coordinate system it is into SRTM30 elevations, the only way to estimate the top of the spillway is to look at the monitoring data. The maximum water level of the reservoir will be at or slightly above the spillway. During flood events it could be several metres above the top of the spillway, so use your discretion.



The primary and secondary axis are automatic. However, they may have to be adjusted to **line up the vertical grid lines between the Capacity and Area axis**. Follow this procedure:

- Start with the Capacity axis and adjust the maximum so that the data fits nicely on the chart and the major gridlines are logical. E.g., 0 -3,500 with intervals of 500 MCM.
- Adjust the Area axis, until the major gridlines are the same as the ones for the Capacity Axis, and the intervals are logical.

- **Adjust the minimum & maximum of the Elevation y-axis**. For Mirani, a good range is from 40-100m. Note that you need to adjust the primary (on the left) and secondary (on the right) y-axis.
- **Adjust the Axis Number format**: They are set to General, so it works for all data ranges. Select the primary x-axis (capacity) | **Format Axis | Axis Options | Number | Category**
- Select Number
- Decimal places: 0
- Select Use 1000 Separator (,)

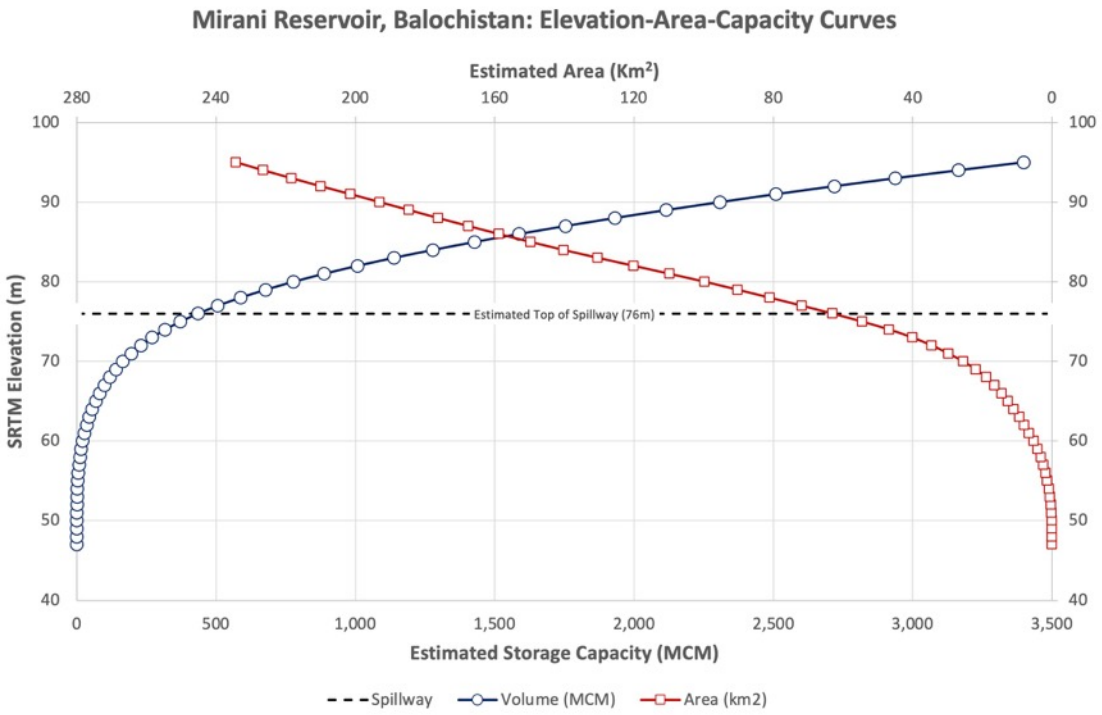


Now we will adjust the dashed line showing the top of the spillway.

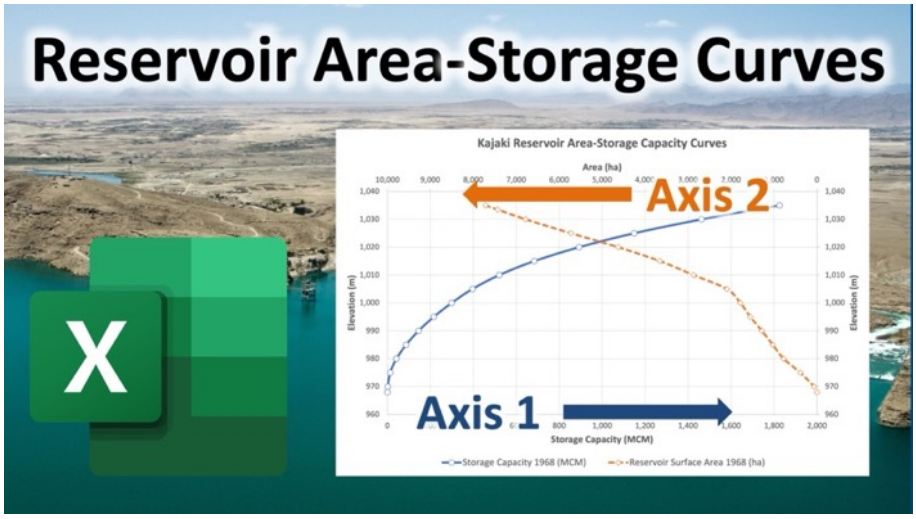
- Go to the **Info sheet | Spillway table**
- Adjust the maximum x-value so that it is the same as the chart. For Mirani it should be 3,500
- Change the y-values to an estimate of the spillway height in SRTM30 elevation. For Mirani use 76.

This is for the Reservoir Curves chart to show the top of the spillway x = range of storage capacity (MCM), y = water level				
x	y	Name		
0	76			**Formulas used**
1750	76	Estimated Top of Spillway (76m)		=E8*0.5
3500	76			=MAX(tbl_Data[Volume (MCM)])
Estimate the top of spillway from monitoring data				

- Have a look at the **Reservoir Curves** chart.
- Note that at the top of the spillway, the reservoir capacity is a little bit over 400 MCM, with a surface area of about 60 km^2.



If you want to learn how to make a chart like the one used for the Reservoir curves, see the YouTube video **Excel: How to add a 2nd Axis to a Chart - Reservoir Elevation-Area-Storage Capacity Curves:** https://youtu.be/vVuxrQbNWDU



5 Download Reservoir Surface Water Area Data from Satellite Remote Sensing

Once you have set up the Elevation – Area – Capacity curves for a reservoir in the MS Excel template, you can use it to monitor the reservoir via remote sensing of the reservoir surface area.

5.1 Satellites for Reservoir Surface Water Area Monitoring

5.1.1 Water Area Monitoring using optical sensors

Satellites using optical sensors to detect water surfaces have the longest data records going back to the 1980s. However, they have one big disadvantage which is that if the reservoir is covered in cloud the water surface cannot be measured. Key Satellites are:

- **Landsat Series (NASA):** Landsat satellites (e.g., Landsat 5 TM, Landsat 7 ETM+, Landsat 8 OLI-TIRS[1]) are widely used due to their moderate spatial resolution with a pixel size of 30m, and long historical archive since 1982. Note that the older satellite data is often less frequent or even missing from some parts of the world especially the more remote areas.
- **Sentinel-2 (ESA):** Sentinel-2 satellites offer higher spatial resolution with 10m for visible and NIR bands and 20m for SWIR, and a shorter revisit time of 5-10 days, providing more detailed and frequent observations.
- **MODerate Resolution Imaging Spectroradiometer (MODIS):** MODIS offers daily coverage, which is an advantage for frequent monitoring, though its resolution is coarser (250m to 500m) compared to Landsat or Sentinel-2.

5.1.2 Water Area Monitoring using Synthetic Aperture Radar (SAR) sensors:

SAR sensors are advantageous because they can penetrate clouds and operate day or night, making them valuable in all weather conditions.

- The key Satellite providing free data is **Sentinel-1 (ESA)** with a resolution of 5 to 20m, and a revisit time of 6 days (two operational satellites) to 12 days (one operational satellite).

5.2 Downloading free Reservoir Surface Water Area Monitoring data

The calculations required to measure a reservoir area are immense, as each pixel of each satellite image needs to be processed. Sometimes there are hundreds of satellite images available which means millions of pixels will need to be processed. There are two options to do this:

- **Process the data in Google Earth Engine** using specialized python scripts. This requires a professional level of programming skills, and although scripts can be copy-pasted, the data analysist needs to understand exactly what is going on or there is a high risk that the analysed data will be meaningless.

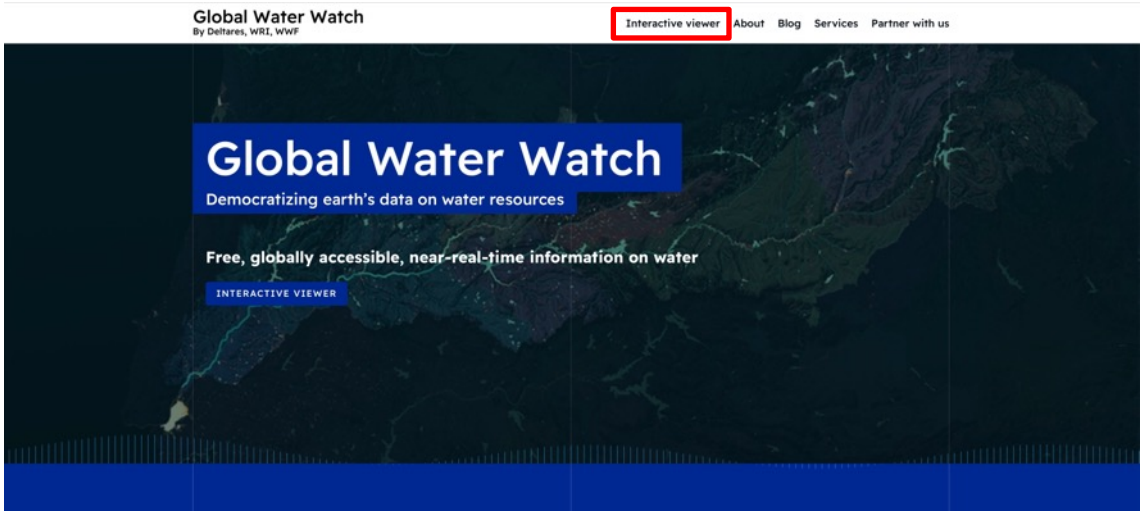
[1] Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS)

- **Download data provided by online Apps:** There are several institutions which provide reservoir area monitoring data. This is the preferred option for downloading reservoir area monitoring data, as the data is controlled by professional organisations. One disadvantage is that not all reservoirs are currently monitored, but their number is increasing all the time.

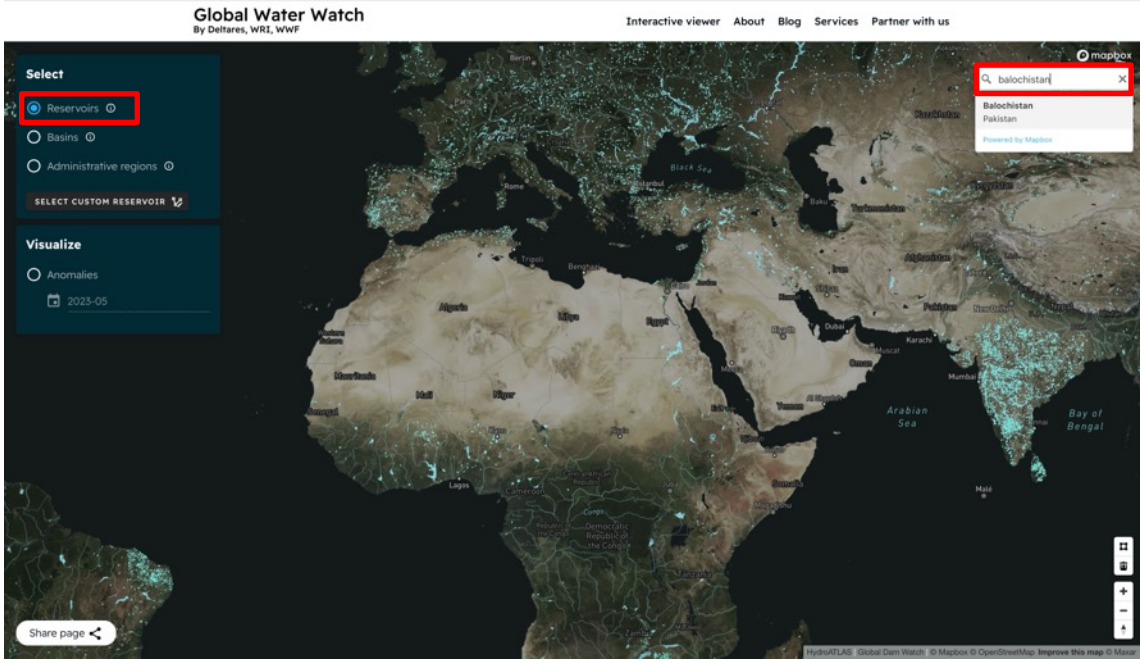
5.3 Download surface water area data from Global Water Watch

The Global Water Watch is a free online source of reservoir surface water area monitoring data supported by Deltares, WRI and WWF so we can expect good quality data from this website.

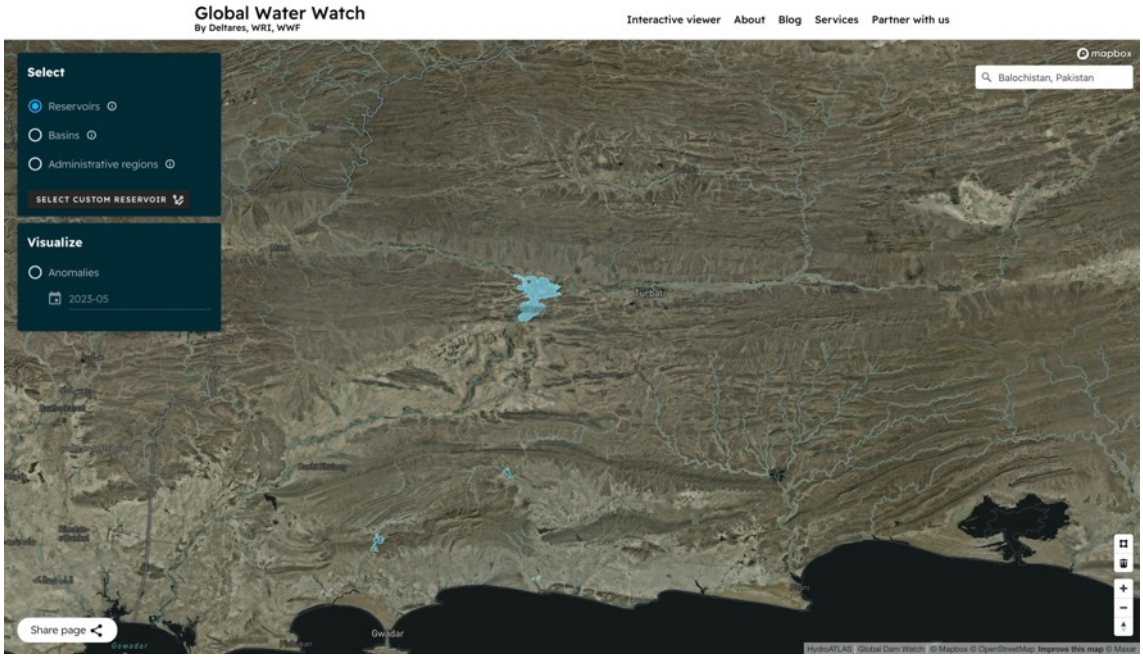
- Go to the Global Water Watch at: **https://www.globalwaterwatch.earth** and Click on the **Interactive Viewer**
- Alternatively go straight to **https://www.globalwaterwatch.earth/map**



- Select **Reservoirs** & use the search bar to **find Balochistan**



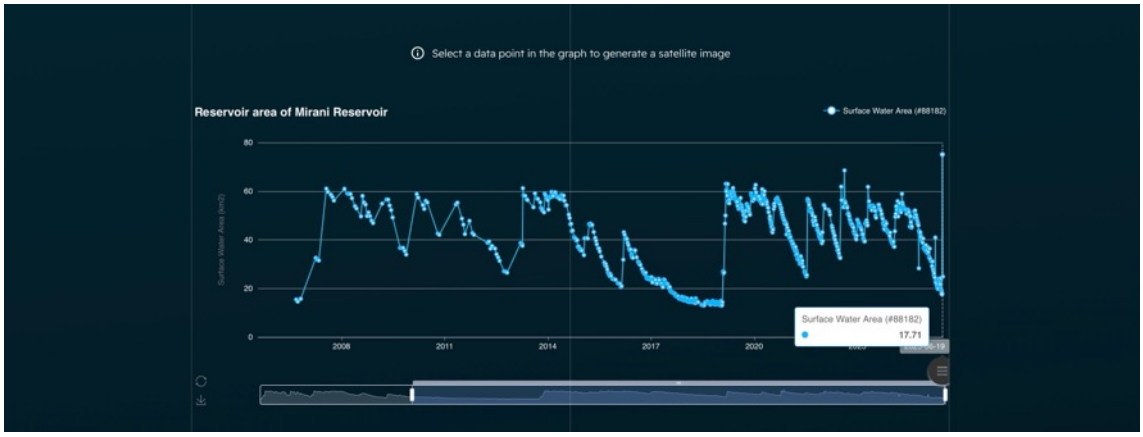
- Zoom in towards the Mirani Dam
- Note that all reservoirs which are being monitored have a pale blue outline



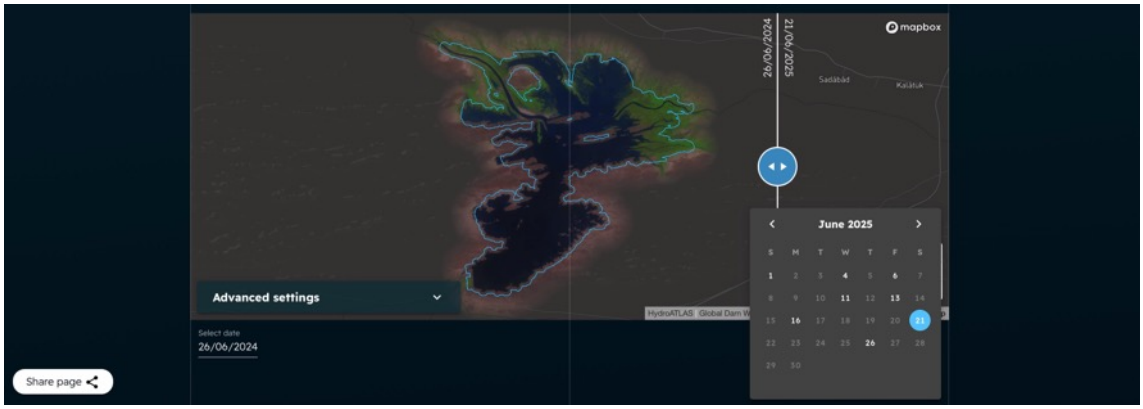
- Click on the **Mirani Reservoir**
- A dedicated page opens which monitors the Mirani Reservoir https://www.globalwaterwatch.earth/reservoir/88182
- The pale blue line represents the maximum flooded area, and you can select the satellite image from the reservoir area by selecting a point on the timeline below.



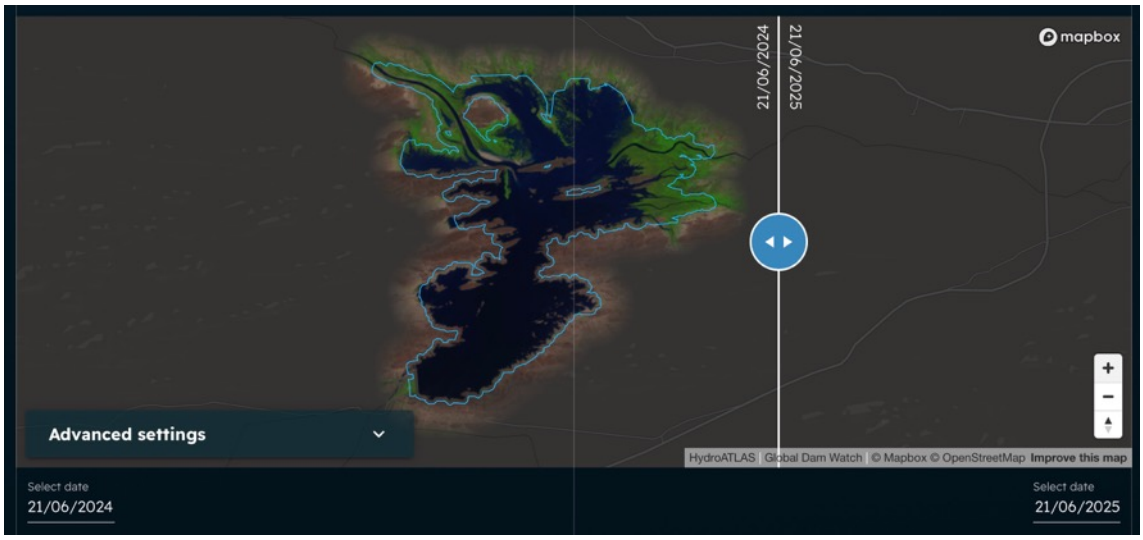
- Scroll down to the graph of the variations in reservoir water area over time since 2007. You can adjust the slider below it to visualize the data available.
- Each point on the line represents a data point. If you click on one of the points it will show you the water area in km2, and the satellite image above will update. The image above is from the 19 June 2025. Note that there appears to be a much larger water area in the next two satellite images, but this is a misconception. During that time the reservoir was covered in clouds, so the water classifier algorithm has overestimated the surface area.

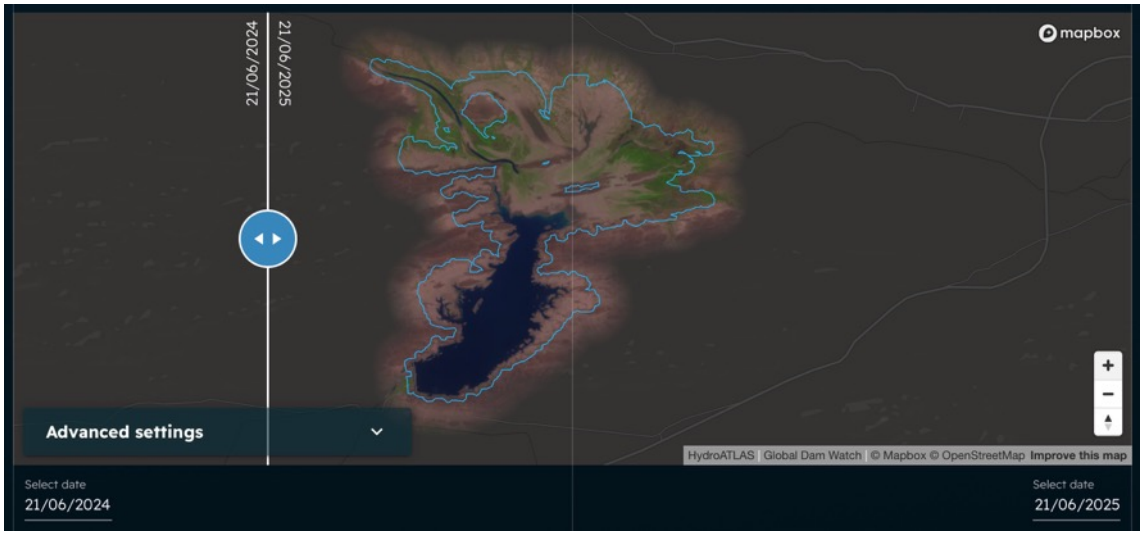


- Scroll down to the next reservoir image which is a comparison map between two satellite acquisition dates.
- **Click on the date under Select date** on the right side and pick the last date where there is no cloud cover. In this case we **choose 21 June 2025**, instead of 26 June 2025.
- **Adjust the date on the left side to 21 June 2024**.

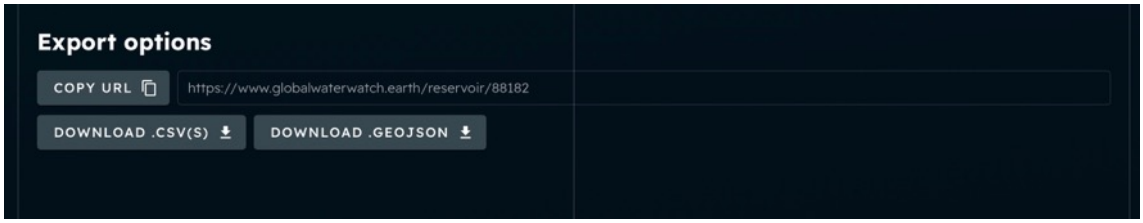


- Slide the slider left & right across the Mirani reservoir to compare water areas between 21 June of 2024 and 2025.
- Note that there is much less water in the reservoir in 2025 and the vegetation in the Northern part of the reservoir area is much less developed.





- Scroll down to the Export options
- Click on **DOWNLOAD .CSV(s)**
- A CSV file of the Mirani Reservoir water area data will be downloaded to your computer. It is called Surface Water Area (#88182).csv, where 88182 refers to the ID number of the Mirani Reservoir.



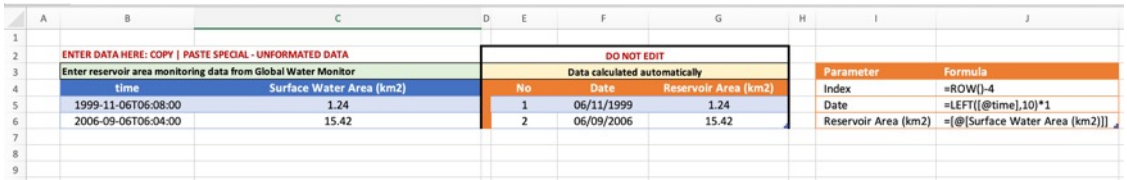
5.4 Process surface water area data in Excel

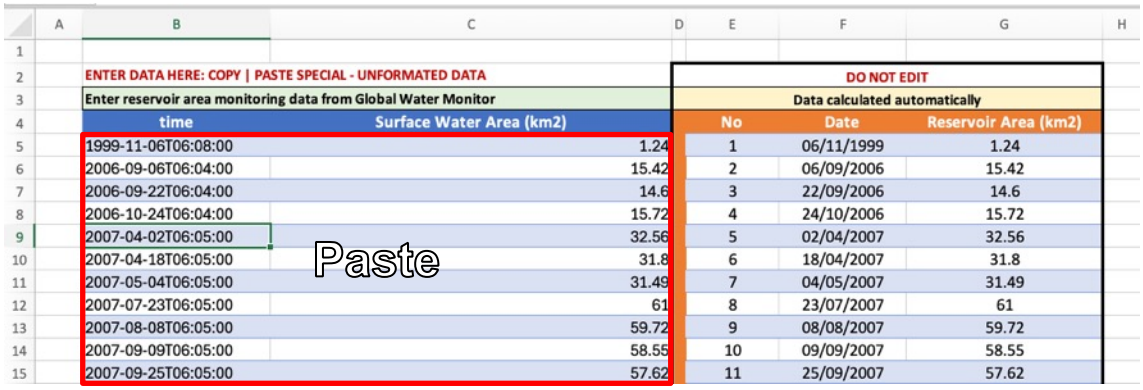
The next step is to process the reservoir surface water area data in the Excel template.

- Open Surface Water Area (#88182).csv and save it as an XLSX file.
- Note that there are two columns:
  - **time** with the format e.g., 1999-11-06T06:08:00, and
  - **Surface Water Area (km2)** with a decimal format e.g., 1.24

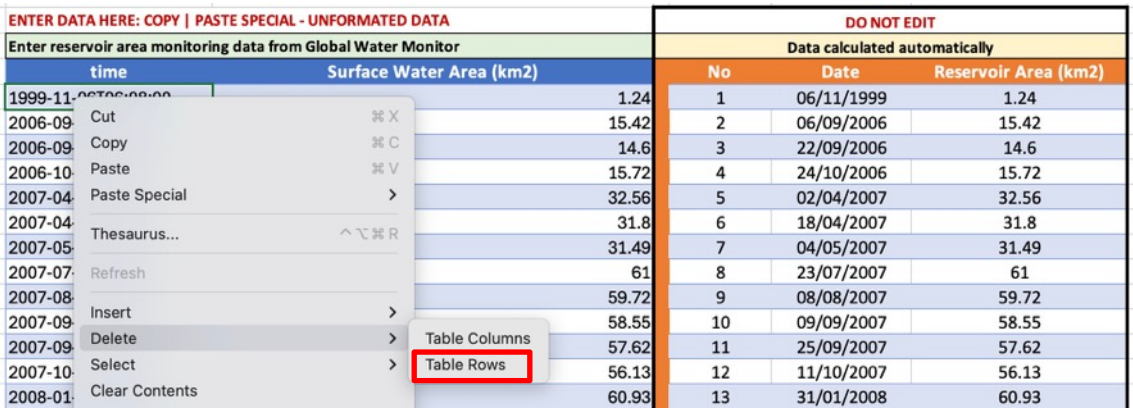
	A	B	C	D	E
1	time	Surface Water Area (km2)			
2	1999-11-06T06:08:00	1.24			
3	2006-09-06T06:04:00	15.42			
4	2006-09-22T06:04:00	14.6			
5	2006-10-24T06:04:00	15.72			
6	2007-04-02T06:05:00	32.56			

- Open Mirani Reservoir Curves V1.xlsx
- Go to the Monitoring sheet
- Copy-paste the **time** & **Surface Water Area (km2)** from the Surface Water Area (#88182) file to Mirani Reservoir Curves V1.xlsx
- Do not touch any of the other columns as they will automatically convert the data into the correct format for Excel.

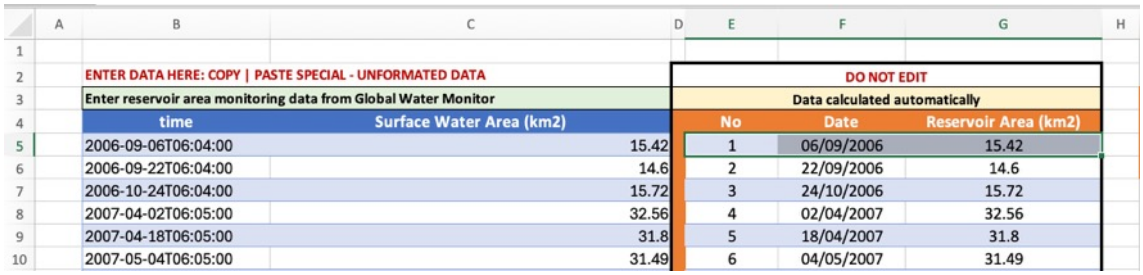




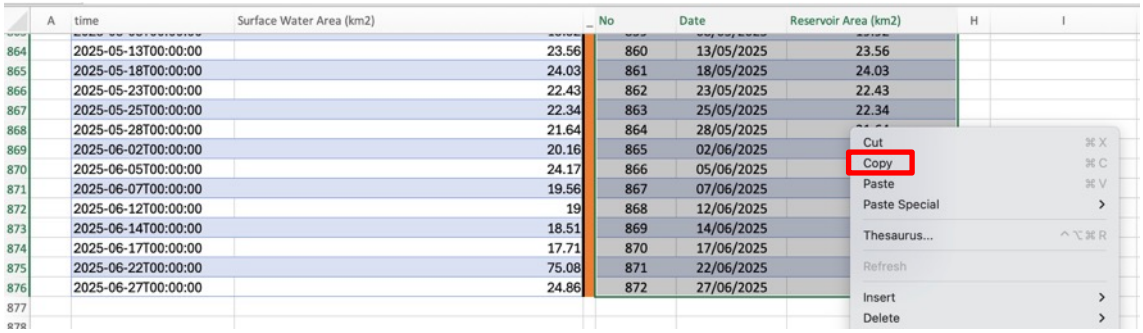
- The first data point from 1999 is a long time before the reservoir was filled, so we will delete it.
- Select the time cell of the 1999 data, **Right-click | Delete | Table Rows**



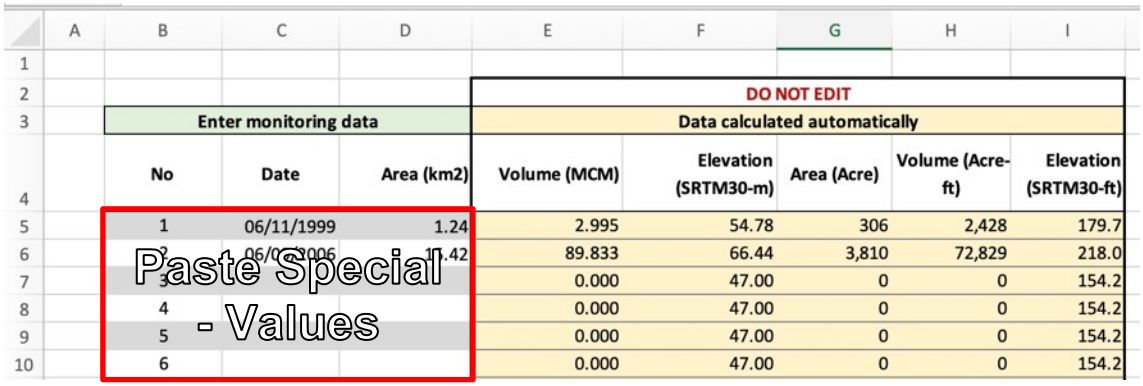
- Select the top row from the calculated data in the orange table to the right.
- **Shift | CRTR | Arrow down** to select all the data in No, Date & Reservoir Area (km2) columns.



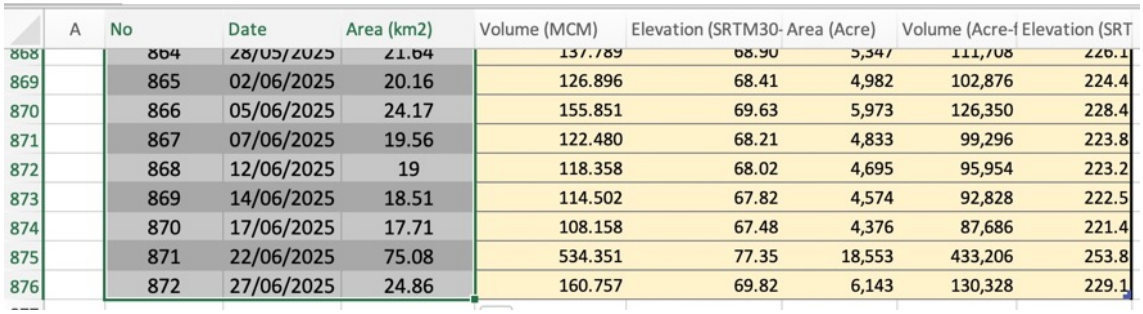
- Check that all the data has been copied down to the bottom of the table.
- **Right-click | Copy**



- Go to the **Calcs** sheet
- In the **left three columns** under the green Enter monitoring data, **Paste Special | Values** the data you have just copied.

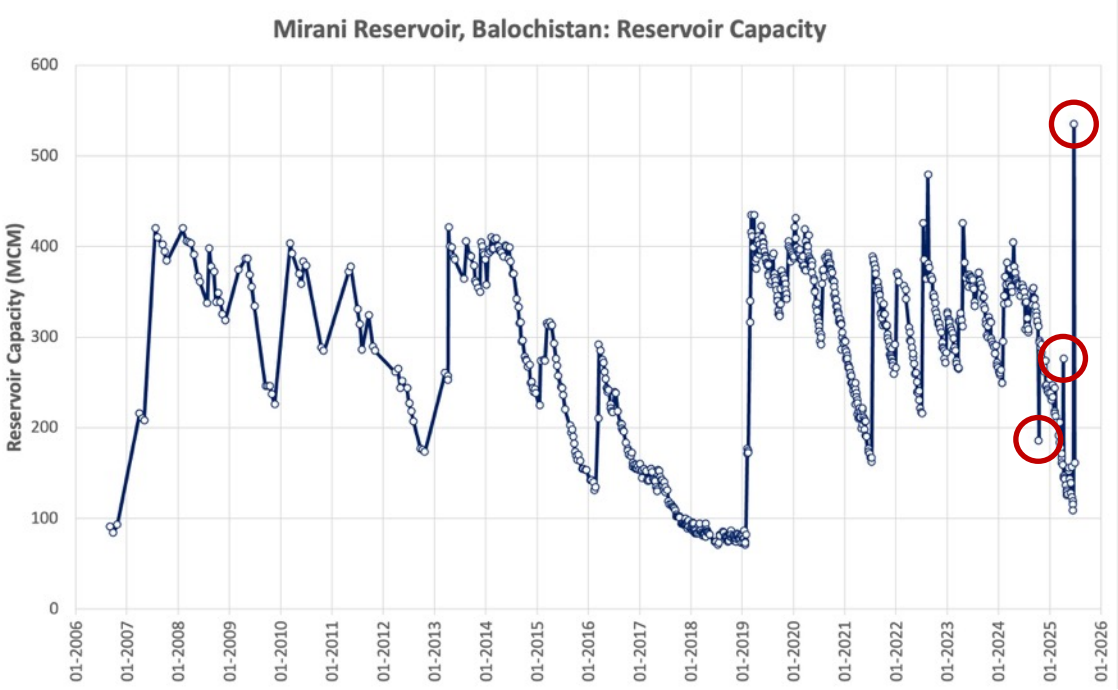


- Check that it has pasted all the way to the bottom. The orange calculated cells should have filled down automatically.

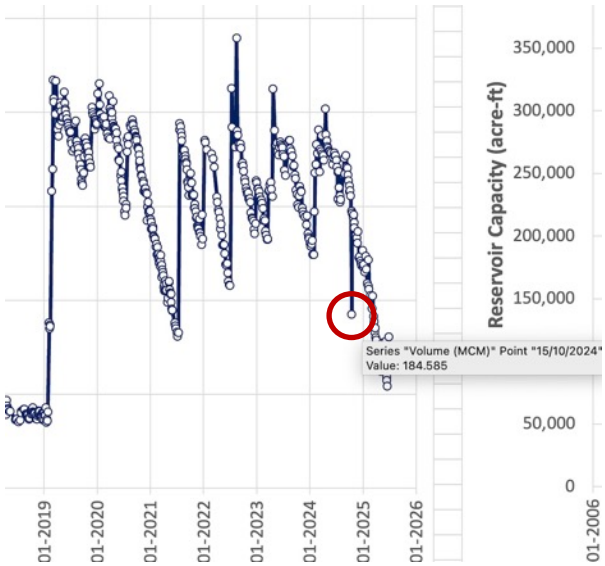


In the calcs sheet you will find six charts which are all linked to the table we have just updated with reservoir surface area monitoring data. These are:

- Reservoir Capacity (MCM)
- Reservoir Capacity (acre-ft)
- Water Surface Water Area (km2)
- Water Surface Water Area (acres)
- Reservoir Water Level Elevation (SRTM30-m)
- Reservoir Water Level Elevation (SRTM30-m)



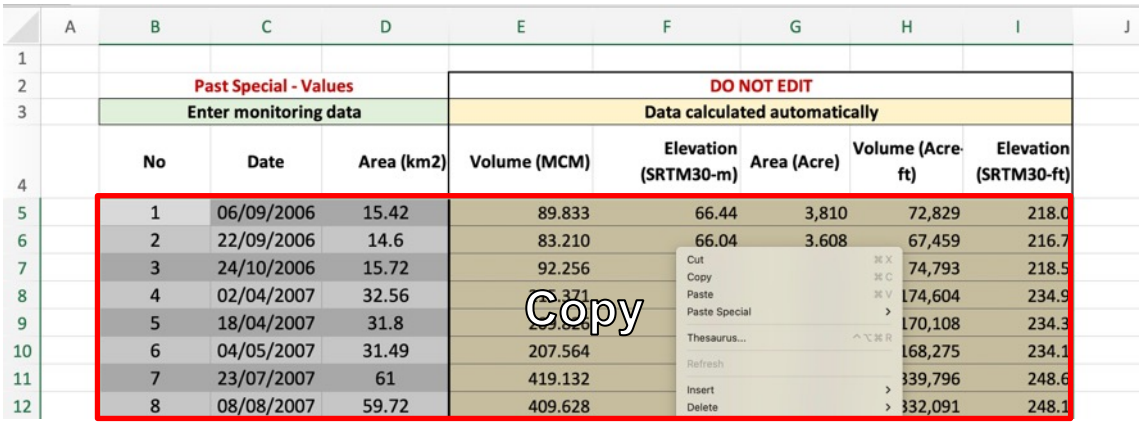
- Hover over a spiky data point & wait until the data and value of that point are displayed in the chart e.g.,:
    - 22/06/2025
    - 08/04/2025
    - 15/10/2024



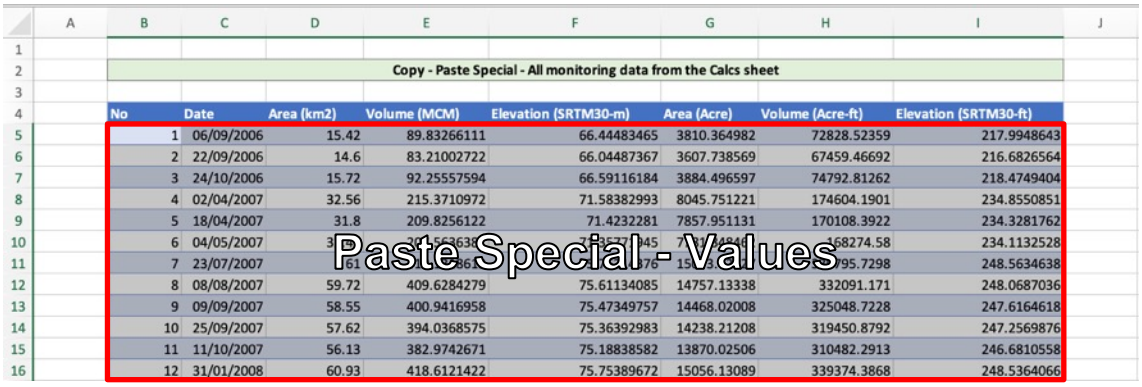
- Check the satellite images in Global Water Watch. Note that the dates may be offset by one day due to differences in time zones. e.g., for 22/06/2025 you need to click on 21/06/2025. **If there are clouds or the images appear blurred delete that data point.**

- Very important: in the Menu bar click on **Data | Refresh All**

- Copy all the data from the Calcs table



- Go to the **Export** sheet
- Select **cell B5**
- **Paste Special | Values**



This table shows all the final quality-controlled monitoring data for the Mirani Reservoir. You can use it to compare with any manually measured data, such as the elevation of the water level in feet. However, remember that there will most likely be an offset with the SRTM30 elevations. For the Mirani dam it is about 40cm.

- Go to the **Annual Chart sheet.** This is a Pivot table & pivot chart showing the minimum, average and maximum annual reservoir capacity in MCM.

	A	B	C	D
1	**Mirani Reservoir, Balochistan: Reservoir Capacity**			
2				
3	**Row Labels**	**Min of Volume (MCM)**	**Average of Volume (MCM)**	**Max of Volume (MCM)**
4	2006	83.2	88.4	92.3
5	2007	207.6	329.9	419.1
6	2008	317.5	368.2	418.6
7	2009	225.4	309.0	386.1
8	2010	283.7	356.5	402.4
9	2011	283.6	321.5	376.3
10	2012	172.1	221.2	264.0
11	2013	251.6	366.6	420.5
12	2014	237.1	340.3	409.0
13	2015	151.9	227.4	315.5
14	2016	129.9	196.8	290.9
15	2017	87.3	120.0	159.0
16	2018	70.3	81.0	94.0
17	2019	69.9	341.8	434.1
18	2020	284.7	360.3	430.6
19	2021	161.1	269.0	387.7
20	2022	215.2	311.7	478.6
21	2023	264.3	320.5	424.6
22	2024	237.0	318.3	403.4
23	2025	108.2	170.0	246.6
24				

